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What can real cause–effect benchmarks resolve? Sampling noise, marginal shortcuts and assumption regimes

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Abstract: Causal-discovery methods are compared by small accuracy gains on a few real benchmarks. We ask what these benchmarks can resolve and why rules that test the conditional structure of a pair rarely beat simple rules. Twelve direction rules were run under one pre-written protocol on eight real pair collections, including four new Causal Chambers collections, and on graph benchmarks. No rule leads across collections, and rule rankings differ between collections. On 102 Tübingen pairs a paired test needs a difference of 19 accuracy points for 80% power, and among ten non-degenerate rules no pairwise difference reaches a raw $p < 0.05$. Unreported conventions move a rule as much: a rounding choice changes a published baseline's weighted accuracy by six points, and the treatment of tied values changes IGCI from 94% to 6% on one collection. Two mechanisms are supported: the chamber and Sachs pairs can be oriented from the marginals alone, and additive-noise rules are near-perfect when the noise scale is independent of the cause and reversed when it depends on it. Attempts to turn this into a rule or a predictive model fail their pre-registered predictions, so the account is partial. We give reporting recommendations.

Keywords: causal discovery, cause–effect pairs, benchmark evaluation, reproducibility, additive noise models

MSC: 62D20, 68T05

1 Introduction

Cause–effect benchmarks drive method development in causal discovery. For two variables the reference is the Tübingen collection [1], 108 pairs of real measurements of which 102 are univariate, and a method is summarised by its accuracy at orienting them; for graphs it is a small number of real networks, most often the protein-signalling data of Sachs et al. [2] with 17 consensus arcs. A new rule is accepted when its number is larger than the numbers of earlier rules in a table. This paper asks how much information such a table carries. It does so by running a panel of direction rules, under one protocol written before the runs, on eight real pair collections and on the graph benchmarks, and by asking two questions of the results: how large a difference the benchmarks can resolve, and why dependence-based rules, which use the structure that causal discovery is built on, so rarely beat much simpler rules on real data.

We use the word *baseline* for rules that read the scale of regression errors or the marginal density of the variables: RECI [3], IGCI [4] and, since 2026, a parameter-free compression score [5]. The *dependence-based* rules are those that test a model of the conditional distribution: additive-noise models [6, 7], location-scale models [8], pairwise likelihood-ratio rules for non-Gaussian linear models [9], and amortised or learned models [10]. On the Tübingen pairs all of them lie between 55% and 67% in our hands, with intervals of ±10 points; on chamber recordings with a known graph the ranking is almost the reverse.

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Contributions.
(1) A same-protocol comparison of twelve pair rules (thirteen with one variant) on eight real collections (Tübingen; six Causal Chambers collections, four of them new; the Sachs pairs), with the stored decision of every rule on every pair, cluster-aware intervals, and a reproduction audit against a published number (Section 11). Reproducing the published accuracy of the compression baseline exactly required one undocumented rounding convention, which by itself moves the weighted headline by six points. (2) A resolution analysis: with 102 pairs a paired test has 80% power only for a difference of about 19 points; of 78 pairwise comparisons two survive a Holm correction and both involve a rule below chance, and among ten non-degenerate rules none of 45 comparisons reaches a raw $p < 0.05$ (Section 5). (3) A multiverse of ten reporting universes that moves the accuracy of a single rule by 11 to 26 points and its rank by up to ten places (Section 6). (4) Evidence on mechanisms, each tested by prediction, by control or by manipulation (Section 9): the chamber and Sachs benchmarks are largely solvable from the marginal distributions alone, validated across recordings and across proteins; additive-noise rules are near-perfect when the residual scale does not depend on the cause and reversed when it does, shown by a matched simulation with a marginal control and a factorial map; two attempts to turn this account into a rule or a predictive model fail with their pre-registered predictions, which bounds the explanation (Sections 9.5, 9.6). (5) An accounting of the attempts to build a better rule during the study (Appendix C) and the pre-registration log of the experiment scripts (Appendix A).

What this paper does not claim.
It does not claim that published methods misreport their results. We audited one published table in detail and reproduced it (Section 11); for the others we do not know the protocol used and run the methods ourselves. It does not claim that dependence-based methods are worse in general: they are among the best rules in the regimes that fit their assumptions and the worst in others, and a benchmark that mixes regimes averages them. It does not claim that the mechanisms we identify are the only ones. The predictive test of Section 9.6 fails, which means that the account is incomplete.

Outline.
Section 2 places the work. Section 3 describes the collections and rules, Section 3.3 the protocol and its provenance. Section 4 gives the cross-collection results. Sections 5–7 quantify resolution and reporting freedom. Section 9 tests mechanisms. Section 10 reports the graph benchmarks. Section 11 is the reproduction audit. Section 13 discusses and recommends; Section 14 lists the limits.

2 Related work

Bivariate direction rules.
The identifiability results for linear non-Gaussian models [11], additive-noise models [6, 7, 12] and post-nonlinear models underlie most pair rules. IGCI [4] rests on the independence between the cause distribution and the mechanism; RECI [3] compares regression errors after scaling to the unit interval; pairwise LiNGAM statistics [9] compare likelihoods of non-Gaussian linear models. LOCI [8] fits location-scale noise models with neural networks and an independence test. SLOPE [13] uses minimum description length with a regression-based model class. Learned approaches train a classifier on simulated pairs [14]; AVICI [10] amortises structure inference over simulated data and is the learned graph method we run.

Evaluation of causal discovery.
Real data with known cause–effect structure are scarce. The Tübingen collection [1] is the standard for pairs; the Causal Chambers [15] provide physical devices with a documented graph and interventional recordings; the Sachs network [2] is a consensus rather than a ground truth. For graphs on simulated data, Reisach et al. [16] showed that the variance structure of simulated additive-noise models can be read by a trivial rule, which then rivals score-based methods; our marginal-only classifier is a pair-level analogue of that observation on real data. A same-hands re-evaluation on the Tübingen pairs, which also introduced the compression baseline we use, is Stienstra [5]; we reproduce its main figure in Section 11.

Recent graph methods.
BF-BIC with BOSS [17] scores nonlinear recursive models with truncated basis functions; SNOE [18] orients edges with pairwise additive-noise likelihoods; topological residual asymmetry [19] compares residual clouds. We implement versions of all three from their descriptions where public code was not available, and say where our constants are our own (Section 3).

Reporting and statistical resolution.
The concern that benchmark comparisons are underpowered and that reporting choices are free parameters is general in machine learning and statistics; we do not review it. What is specific here is that the quantities are measured on causal-discovery benchmarks, with exact paired tests on stored decisions and an enumerated multiverse of choices that have each been used for pairs.

3 Collections, rules and protocol

3.1 Pair collections

Table 1 lists the eight collections (the seventh, the walks of all 19 untouched recordings, contains the five recordings of the walk collection and fourteen more). A *pair* is an ordered pair of variables with a known direction; the cause is the first variable of an edge of the documented graph. Two features of the table matter for every later statement. First, only Tübingen is a collection of unrelated mechanisms; the six others consist of pairs taken from one or two devices, so the number of *distinct edges* is much smaller than the number of pair-runs (47 distinct edges behind the 94 i.i.d. chamber pair-runs, at most 65 distinct cause–effect names behind the 122 walk pairs, 8 behind the Bernoulli collection, 7 behind the changepoint collection and 4 behind the Malus collection, whose pairs share their cause variable). Intervals and tests that treat pairs as independent therefore overstate the information; Appendix Table 16 gives cluster-bootstrap intervals (clusters: distinct edge, or recording for the walks). Second, in the chamber collections the cause is an actuator whose value the experimenter sets, which gives it a recognisable marginal law (uniform on a grid of levels, piecewise constant, a random walk) while the effects are sensors.

Table 2 describes the collections with the quantities used later: the median number of rows, the median absolute Spearman correlation, the median residual heteroscedasticity h of Section 9.2 (true direction), the median excess kurtosis of the cause and of the effect, the median number of distinct values of the cause, and the share of pairs with many ties (fewer distinct values than 90% of the rows) in either variable before any jitter (Tübingen is reported after dequantisation, hence zero). Figure 1 shows one randomly chosen pair per collection.

Table 1. The eight real pair collections. "Pairs": pair-runs scored; "edges": distinct edges of the documented graph; "units": independent recordings or experiments. Rows: rows per pair after the cap of 2000 random rows (Tübingen: at most 2000 rows, dequantised; chamber collections: raw values).

Collection	Pairs	Edges	Units	Rows	Source and cause
Tübingen	102	102	102	≤ 2000	Univariate pairs of the CEP benchmark [1]; official weights
Chambers iid	94	47	2	2000	Wind and light tunnel with i.i.d. actuator settings [15]; two seeds per edge
Walks, all 19	486	≤ 65	19	2000	All untouched recordings (wind-tunnel random-walk actuators, the pressure-control recording, light-tunnel walks); every edge of each documented graph
Chambers walks	122	≤ 65	5	2000	First recording of each type of random-walk experiments (actuators are random walks); the same cause–effect names recur across the five recordings
Sachs	17	17	1	853	Consensus arcs among 11 proteins, log scale [2]
Bernoulli	16	8	1	2000	Wind tunnel, `random_loads_both`; the cause `load_in` takes 101 levels (steps of 0.01), so after subsampling it has about 20 rows per value; cause always `load_in`
Changepoints	70	7	10	2000	Wind tunnel, piecewise-constant actuator (change points); cause always `load_in`
Malus	48	4	12	1000	Light tunnel with polarisers, Malus's law; causes `pol_1`, `pol_2`

3.2 Rules

Twelve rules, all run by us. Each returns a score whose sign gives the direction; a score of exactly zero counts as half correct. Settings were fixed in the script before the run (Appendix A) and none was tuned on a collection in which it is scored. *Tie handling:* every rule receives the raw arrays (ties kept), as for the chambers in the first experiments; Tübingen arrays are dequantised (experiment 14) and have no ties. Giving the rules a copy with a 10^{-6} jitter on tied values reverses IGCI (Section 12) and breaks TRA's spline fit, so jitter is used only in the sensitivity analysis of Section 12 and for the simulations.

- **IGCI** [4]: entropy estimator with a uniform reference. **RECI** [3]: the direction with the smaller regression error of a spline fit after min–max scaling. Both are short re-implementations (`sprint_cd/pair_baselines.py`).
- **ANM-HSIC** [6]: additive-noise model with a spline regression and an HSIC independence test; **ANM-GP**: the `causal-learn` Gaussian-process version on 400 rows; **RESIT** (`lingam`): random-forest regression of depth 4 with $\alpha = 0.01$.
- **Pairwise LiNGAM** [9]: first-order likelihood-ratio statistic.
- **LOCI** [8]: the authors' code in its default setting (neural networks, 5000 steps, HSIC test, at most 1000 rows).
- **Compression** [5]: quantise both variables to 256 levels, sort one by the other (stable), take the first difference as `int8`, compress with bzip2 (level 9), and compare the two normalised lengths. Two variants: our first implementation (`floor`$(256q)$, rows capped at 2000) and the *faithful* variant that reproduces the published figure (`round`$(255q)$, all rows thinned linearly above 8000, raw values; Section 11).
- **TRA** [19]: cross-fitted smoothing-spline residuals, a rank-uniform residual cloud and the zero-dimensional persistence of its minimum spanning tree in a window; the constants $\kappa = 1$, $c_\beta = 4$, five folds, at most 1000 rows are ours because the preprint does not fix them. It returns no value when

Table 2. Descriptive diagnostics of the seven original collections (medians over pairs unless stated; the 486-pair walk collection has the same kind of pairs as the walk collection).

Collection	Pairs	n	$\lvert\rho_S\rvert$	h	kurt. cause	kurt. effect	distinct cause values	ties
Tübingen	102	392	0.50	0.18	-0.2	0.2	392	0.00
Chambers iid	94	2000	0.27	0.02	-1.2	-0.4	86	1.00
Chambers walks	122	2000	0.26	0.08	-1.2	-0.6	255	1.00
Sachs	17	853	0.08	0.08	3.1	2.3	276	1.00
Bernoulli	16	2000	0.53	0.18	-1.2	-1.2	101	1.00
Changepoints	70	1983	0.45	0.22	-0.7	-0.1	4	1.00
Malus	48	1000	0.02	0.02	-1.2	-1.4	732	1.00

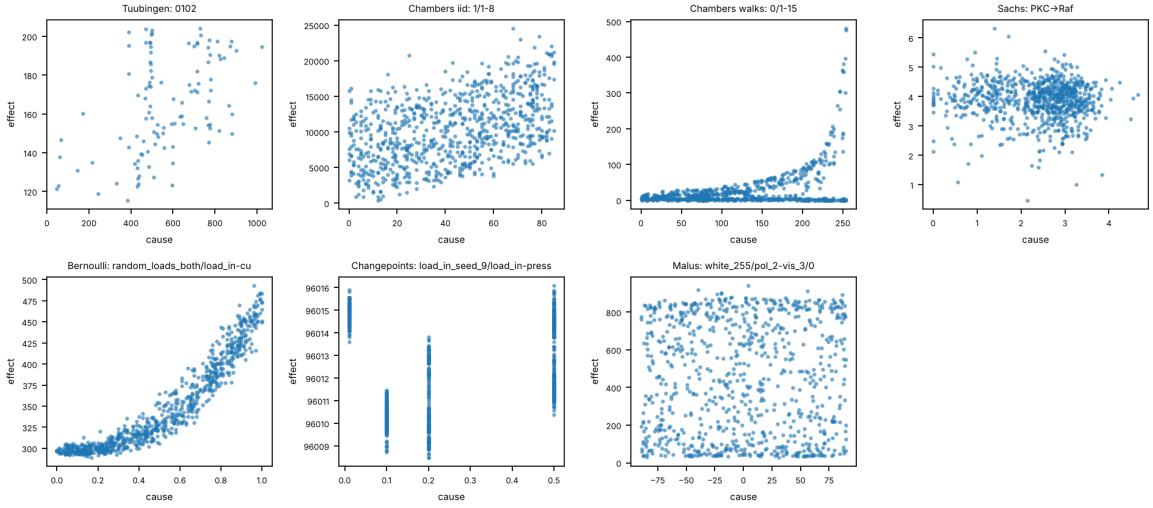


Figure 1. One randomly chosen pair (seed 7) from each collection; 800 random rows, cause on the horizontal axis.

the spline fit is ill-posed (cause with many tied values), which excludes it from the collections with binary or piecewise-constant causes.

– **AVICI** [10]: the pretrained `scm-v0` model on the two-variable input, score $P(x \to y) - P(y \to x)$ (run in a separate environment with pinned JAX).

– **PDM**: a gradient-boosting classifier trained on simulated pairs with 32 features (ours; Appendix C).

Reference implementations of IGCI, RECI, CDS, ANM and BivariateFit from the `cdt` package (v0.6.0, defaults) were also run to check ours (experiment 55; Section 12). Not run: PNL (880 s per pair on 400 rows), CAM and SCORE (no package available offline), SLOPE (code not run), and rules that need the authors' trained weights.

3.3 Protocol and provenance

Every evaluation script was committed before it was run; Appendix A lists the 46 experiment scripts (numbers 17 to 60; 31 and 34 have two parts) with the commit that fixed them. Stored per-pair decisions are in `results/`, and every number in this paper is computed from them by the scripts in `scripts/`. Three analyses were designed after seeing per-pair results and are labelled as such where they appear: the multiverse and subset analyses of Sections 6–7, the terciles of Table 7, and the revised protocol of Section 9.6. The study includes departures that we report rather than hide: our first compression implementation was wrong in a detail that the published text does not fix; the first run of the marginal-only classifier

Table 3. Accuracy (fraction of pairs oriented correctly; exact ties half) of twelve rules on eight collections. All rules see the raw arrays, except that Tübingen is dequantised (Section 3.2). "–": not run, or no value for most pairs (TRA on the changepoint collection; PDM and the first compression implementation only on the original collections). "Walks (19)" contains all 19 untouched recordings (486 pairs; the 122 pairs of "walks" come from five of them). The new collections have 8, 7 and 4 distinct edges; intervals that account for clusters are in Appendix Table 16.

Rule	Tüb. (102)	iid (94)	walks (122)	Walks (19: 486)	Sachs (17)	Bern. (16)	Chgpt. (70)	Malus (48)
IGCI	0.56	0.94	0.89	0.96	0.65	1.00	1.00	0.08
RECI	0.61	0.98	0.94	0.83	0.53	0.25	1.00	0.56
ANM-HSIC	0.61	0.72	0.63	0.43	0.59	0.31	0.57	0.31
Pairwise LiNGAM	0.61	0.78	0.71	0.67	0.53	0.88	0.43	0.40
PDM	0.40	0.93	0.84	–	–	–	–	–
LOCI	0.65	0.77	0.68	–	0.47	0.56	0.80	0.38
ANM, Gaussian process	0.55	0.77	0.67	–	0.56	0.47	0.54	0.33
Compression, first impl.	0.61	0.30	0.61	–	0.76	–	–	–
Compression, faithful	0.67	0.38	0.52	0.50	0.65	0.50	1.00	0.60
TRA	0.65	0.86	0.61	0.56	0.59	0.62	–	0.54
AVICI	0.59	0.90	0.56	0.63	0.53	0.62	0.40	0.02
RESIT	0.61	0.71	0.65	0.48	0.47	0.25	0.53	0.50

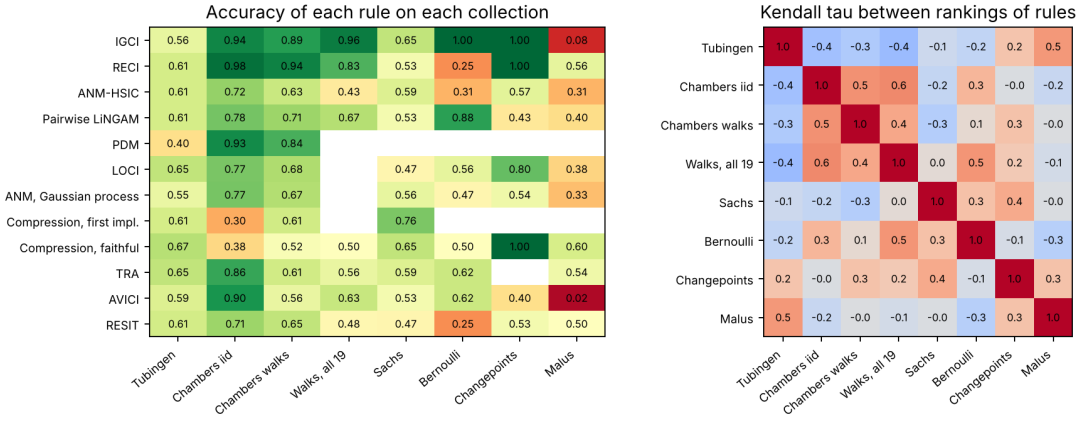


Figure 2. Left: accuracy of each rule on each collection. Right: Kendall rank correlation between the orderings of the rules on two collections (rules run on both; at least five).

leaked information between training and test variables; one pre-registered test (Section 9.6) was invalid as implemented and was rerun in a revised form, with both runs reported.

4 No rule leads across collections

Table 3 and Figure 2 give the accuracy of every rule on every collection. Three observations organise the rest of the paper.

(1) The leader changes with the collection.
The best rule is faithful compression on Tübingen (66.7%; LOCI and TRA at 64.7 and 64.6%) and on the Malus collection (60.0%), RECI on the two chamber collections with independent or random-walk actuators (97.9% and 94.3%), IGCI on all 19 untouched walk recordings (95.7%; RECI 82.9%) and on the

Bernoulli collection (100%), our first compression implementation on the 17 Sachs pairs (76.5%; the faithful one 64.7%), and IGCI, RECI and faithful compression together on the changepoint collection (100%). Four of the twelve rules lead at least one collection (counting the two compression implementations separately; our first implementation leads Sachs by 13 against 11 of 17 pairs for the faithful one, and the faithful one was chosen to match a published figure, so its leads on Tübingen and Malus partly reflect that choice). IGCI is above 70% on five collections and at 8.3% on Malus and 55.9% on Tübingen; RECI is above 70% on four; no other rule is above 70% on more than two.

(2) Rules that agree on one collection can disagree on the next.

IGCI orients the Bernoulli and the changepoint pairs correctly in every case (16/16, 70/70) and the Malus pairs wrongly in 44 of 48. RECI, best or tied-best on three collections, is at 25% on the Bernoulli pairs. The pretrained AVICI model, run on two-variable inputs, is at 90.4% on the i.i.d. chambers and at 2.1% on Malus, where it prefers the reverse direction for 47 of the 48 pairs. Faithful compression is at 100% on the changepoint pairs, 60.0% on Malus and 38.3% on the i.i.d. chambers. The additive-noise rules are never the best: ANM-HSIC is at 43.4% on the 19 walk recordings (cluster interval $[38, 50]$), RESIT at 47.5%. With the recordings as the clusters the ranking of the leaders on the walks is not in doubt: IGCI 95.7% ($[91, 99]$), RECI 82.9% ($[79, 87]$), pairwise LiNGAM 67.1%, AVICI 63.4%, TRA 55.6%, faithful compression 50.2%, RESIT 47.5%, ANM-HSIC 43.4%; the between-recording standard deviation of a rule's accuracy is 0.08 to 0.15.

(3) Rankings of rules do not clearly transfer.

The Kendall correlation between the ordering of the rules on Tübingen and on the i.i.d. chambers is -0.38, against the walk pairs -0.30 (-0.42 for all 19 recordings), against Sachs -0.12 and Bernoulli -0.19; it is positive against the changepoints ($+0.24$) and Malus ($+0.51$). The chamber collections agree with each other (i.i.d. against walks $+0.53$, against all 19 recordings $+0.57$). With a permutation test over the rules none of the Tübingen correlations is distinguishable from zero ($p = 0.06$ to 0.68): the Tübingen accuracies span about twelve points with intervals of ± 0.10, so the ordering of the rules on that collection is mostly noise. The full matrix with permutation p-values is in Appendix Table 20. What it does not show is a shared ranking: a rule that leads Tübingen says little, in either direction, about a collection from a device with a documented graph. The two compression variants and PDM are counted as rules, and the i.i.d. and walk collections come from the same devices.

Cautions.

The cluster-bootstrap intervals of Appendix Table 16 (clusters: 47 edges for the i.i.d. chambers, 5 and 19 recordings for the two walk collections, 8, 7 and 4 edges) are close to the exact intervals for the i.i.d. chambers (RECI 0.98 [0.94, 1.00]) and for Tübingen but are degenerate when a rule is right on every cluster (RECI on the changepoints, 1.00 [1.00, 1.00]) and unreliable with four to eight clusters; their purpose is to show how few independent units the new collections contain. The Bernoulli, changepoint and Malus collections consist of 8, 7 and 4 distinct edges, repeated across seeds or experiments, with the same cause variable (`load_in`, or the two polarisers); they are small, correlated sets of regimes, not samples of mechanisms, and we use them as such. Sachs has 17 pairs. The Tübingen collection mixes pairs from several fields and its weights downweight groups of similar pairs; unweighted accuracy is used here unless stated. IGCI's results depend on how ties are handled (Section 12); the numbers here are for the raw arrays.

5 Sampling noise

Table 4 gives the accuracies on Tübingen with exact Clopper–Pearson intervals. The half-width is about 0.10 for every rule, and the intervals treat the pairs as independent, which they are not exactly (the official weights downweight groups of similar pairs; unweighted accuracy is used). All rules are evaluated on the same pairs, so the right comparison between two rules is the exact McNemar test on the pairs on which exactly one is correct; exact ties (a score of zero) and pairs on which a rule returned no value (TRA, 20 pairs) are left out of the discordant counts, so TRA is compared on a different sample.

Two families of comparisons.
The first family has the twelve rules of Table 3 plus our first-implementation compression with raw rows: 78 comparisons. Twelve have a raw $p < 0.05$ and two survive a Holm correction (smallest adjusted $p = 0.015$); both involve the learned classifier PDM, which is below chance (PDM against faithful compression and against LOCI; 11 of the 12 raw-significant comparisons involve PDM), so they say nothing about the other rules. The second family fixes the ten rules that are not degenerate and not duplicated (IGCI, RECI, ANM-HSIC, pairwise LiNGAM, LOCI, ANM-GP, faithful compression, TRA, AVICI, RESIT; without PDM and without the other compression variants): 45 comparisons, *none* with a raw $p < 0.05$ (smallest raw $p = 0.11$, ANM-GP against RESIT). For the best rule, faithful compression (66.7%), the tests against RECI, LOCI, IGCI, ANM-GP and AVICI have $p = 0.39$, 0.88, 0.14, 0.20 and 0.35 (discordant pairs 20/14, 23/21, 29/18, 24/15, 32/24).

Detectable difference.
With the observed rate of discordant pairs (42%, computed on pairs where both rules have a value and neither ties; averaged over the 78 comparisons), a simulated paired comparison on 102 pairs reaches 80% power for a difference in accuracy of 0.19 at level 0.05 and of 0.27 with a Bonferroni correction over the 78 comparisons. The simulation draws the number of discordant pairs from the observed rate, splits them with the alternative probability, uses 1000 replications per step of 0.01, and treats pairs as independent; the figure is approximate and applies to this discordance rate. Differences of two to five points, which separate most of the rules in recent papers, are far below it: the collection cannot resolve them, whatever the rules.

Graph benchmarks.
The Sachs network has 17 consensus arcs. For 22 method configurations (Section 10), 16 have a 95% bootstrap interval (resampling the 55 variable pairs) that contains the best F_1 (0.67), 7 have a point estimate within 0.10 of it, and of 231 pairs of configurations 62 differ significantly, the smallest significant difference in F_1 being 0.17 (some pairs are the same method at two settings, and resampled variable pairs share nodes, so these are rough). One edge changes recall by 0.06.

6 A multiverse of reporting choices

A reader of a pair-rule paper meets choices that are each reasonable and that each change the number: whether to use the benchmark's official weights; whether to restrict to pairs with enough rows; whether to report accuracy on all pairs or only on the pairs on which the rule is most confident (decided-only accuracy, as for a rule that abstains). We cross them: official weights or none; all pairs or only pairs with at least 250, 500 or 1000 rows; accuracy on the 100, 75, 50 or 25% of the pairs with the largest absolute score. This gives 32 universes. The coverage is taken exactly (top k by |score|, ties broken at random with a fixed seed); the score of several rules is not a calibrated confidence, so this is the operation a reader could apply,

Table 4. Rules on the 102 Tübingen pairs: unweighted accuracy with a 95% interval (TRA: the 82 pairs on which it returned a value) and the multiverse summary of Sections 6–7. "Range": minimum and maximum accuracy over the ten reporting universes with at least 30 pairs per rule; "Rank": best and worst rank; "First": universes in which the rule is first (ties share the rank); ">RECI": universes in which it is above RECI; "Best of 10": expected best accuracy after trying ten random subsets of 50 pairs.

Rule	Accuracy [interval]	Range	Rank	First	>RECI	Best of 10
IGCI	0.559 [0.46, 0.66]	0.47–0.71	4–12	0	0	0.636
RECI	0.608 [0.51, 0.70]	0.61–0.87	1–6	1	0	0.682
ANM-HSIC	0.608 [0.51, 0.70]	0.57–0.70	3–13	0	1	0.684
Pairwise LiNGAM	0.608 [0.51, 0.70]	0.54–0.74	5–13	0	0	0.682
PDM	0.402 [0.31, 0.50]	0.40–0.61	11–13	0	0	0.479
LOCI	0.647 [0.55, 0.74]	0.63–0.77	1–10	1	2	0.721
ANM, Gaussian process	0.554 [0.45, 0.65]	0.55–0.75	3–12	0	0	0.629
Compression, first impl.	0.608 [0.51, 0.70]	0.61–0.79	3–8	0	0	0.684
Compression, first impl., raw rows	0.647 [0.55, 0.74]	0.64–0.76	2–9	0	2	0.720
TRA	0.646 [0.53, 0.75]	0.65–0.76	2–7	0	4	0.730
Compression, faithful	0.667 [0.57, 0.76]	0.67–0.88	1–3	8	9	0.739
AVICI	0.588 [0.49, 0.68]	0.54–0.65	6–12	0	0	0.666
RESIT	0.613 [0.52, 0.71]	0.61–0.76	4–9	0	1	0.687

not each rule's own abstention. A universe is kept only if every rule is evaluated on at least 30 pairs: 10 of 32 qualify (all pairs, or pairs with ≥ 250 rows; coverage 100, 75, and for all pairs 50%; each with and without weights). The other 22 (for example the 25% of the 36 pairs with at least 1000 rows, which is nine pairs) cannot be reported. A first run kept them, and the ranges were far wider (0.27 to 0.75 for IGCI, up to 1.00 for RECI); an accuracy on nine pairs is not a defensible report, and we dropped them on review.

Results.

In the ten universes the accuracy of a single rule still moves by 11 to 26 points (Table 4): 0.47 to 0.71 for IGCI, 0.61 to 0.87 for RECI, 0.67 to 0.88 for the faithful compression, 0.63 to 0.77 for LOCI. Ranks move by up to ten places (ANM-HSIC from third to thirteenth, pairwise LiNGAM from fifth to thirteenth). Three rules are first in at least one universe: faithful compression in 8, RECI and LOCI in 1 each; the faithful compression is above RECI in 9 universes and no other rule is above RECI in more than four. The first counts are over universes, not within one universe, ties share the first rank, and they depend on which variants of a rule are in the panel: the faithful compression convention was selected because it reproduces a published figure (Section 11), and under the other conventions its unweighted accuracy ranges from 60.8% to 69.6%, so its lead in 8 of the 10 universes is partly a consequence of that selection. TRA returned no value on 20 pairs; these are excluded from its accuracy (82 pairs) and from its coverage ranking.

Single choices can be large. The official weights move RECI from 60.8% to 70.9%, faithful compression from 66.7% to 74.7%, and our first compression implementation from 60.8% to 68.0%, while LOCI goes from 64.7% to 63.9%. The quantisation convention of the compression rule (Section 11) moves the weighted accuracy between 68.8 and 74.7%.

7 Selective subsets

Pair benchmarks are often evaluated on a subset of pairs (a given variable type or sample size, or pairs where the other rules disagree). We drew 10000 random subsets of 50 of the 102 pairs. The accuracy of a rule on a subset has a standard deviation of about 0.05. For the eight rules whose full-sample accuracy is within 0.04 of RECI the probability that a rule beats RECI on a subset lies between 0.36 and 0.72, so most

subsets would support either order; for faithful compression, 0.059 above RECI, it is 0.80. The expected best accuracy over m independently drawn subsets, which is what a researcher reports after trying m subsets and keeping the best, grows with m: RECI 0.607 ($m = 1$), 0.649 (3), 0.682 (10); LOCI 0.647, 0.687, 0.721; faithful compression 0.666, 0.706, 0.739. The inflation at $m = 10$ is seven to eight points for every rule, which leaves the order of the rules unchanged; what it changes is the claim of an absolute level. Each subset is half of the same 102 pairs, so this measures selection noise about a known full-sample accuracy, not how a researcher actually chooses subsets (by type or size, not at random).

8 A protocol multiverse on Tübingen

Section 6 varied how a result is reported. Here we vary how the data are prepared, for the same rule and the same pairs (experiment 61): the number of rows (250, 500, 1000, 2000 or all, drawn at random), whether tied values are dequantised, and the seed (which changes the subsample and the rule's own randomness): 50 configurations. Table 5 gives the range of each rule's accuracy and the range of the marginal means of each factor.

The same rule on the same pairs moves by 7 to 22 points unweighted and 8 to 23 points weighted: IGCI with a uniform reference from 49.0 to 65.7%, with cdt's Gaussian reference from 40.2 to 62.7% (the dequantisation alone: 52.4% against 63.8% on average for ours, so on raw Tübingen arrays IGCI is better than RECI, and with dequantisation worse), RECI weighted from 58.4 to 72.5%, faithful compression weighted from 59.9 to 78.4%, CDS unweighted from 53.9 to 65.7%. The number of rows matters little on average (a few points) but through the random subsample the seed alone moves RECI, LiNGAM and ANM-HSIC by two to three points. The range of a single rule across protocol choices is as large as the range between the twelve rules of Table 3 on Tübingen (40 to 67%, or 55 to 67% without the classifier below chance), and larger than the 2 to 5 points by which methods are usually separated. A comparison between two rules is therefore informative only under one identical protocol, run with the same preprocessing and several seeds; a comparison against a number quoted from a paper is not.

9 Why dependence-based rules do not lead: mechanisms

Sections 4–7 say that the benchmarks are small and that no rule leads across them. They do not say why the rules that test the conditional structure of the pair do not do better. We tested three explanations, each with a prediction fixed before the run (experiments 49–52, 56, 57), and report the ones that failed.

9.1 The benchmarks can be solved from the marginals

Hypothesis. On the real collections, a classifier that never sees the joint distribution of (x, y), only the two marginal distributions, orients the pairs about as well as the best rule. If so, the benchmark rewards a rule's sensitivity to marginal asymmetry between cause and effect, a property that conditional-structure rules do not use.

Classifier. Each variable is scaled to $[0, 1]$ and described by five numbers: skewness, excess kurtosis, Vasicek spacing entropy ($m = \sqrt{n}$), the logarithm of the fraction of unique values, and the 1–99% range. The pair feature is the difference of the two vectors; both orientations enter training with labels 1 and 0; the model is a standardised logistic regression. The first protocol (experiment 49) uses five-fold cross-validation with folds over pairs (50 repetitions): 59.0% on Tübingen, 100% on the i.i.d. chamber pairs, 92.6% on the walk pairs and 89.8% on the Sachs pairs.

Table 5. Tübingen, 50 protocol configurations (five row caps × dequantisation or not × five seeds). Range of the unweighted and of the official-weight accuracy over all configurations, and the range (in points) between the best and the worst level of each factor, averaged over the others.

Rule	unweighted	weighted	rows	dequant.	seed
IGCI (ours, uniform ref.)	0.49–0.66	0.57–0.68	1	11	1
RECI	0.56–0.67	0.58–0.73	2	1	2
ANM-HSIC	0.54–0.61	0.52–0.60	3	0	2
Pairwise LiNGAM	0.56–0.64	0.58–0.67	2	0	3
Compression, faithful	0.61–0.71	0.60–0.78	2	1	1
IGCI (cdt, Gaussian ref.)	0.40–0.63	0.40–0.64	1	18	0
RECI (cdt)	0.60–0.68	0.66–0.79	2	2	1
CDS (cdt)	0.54–0.66	0.50–0.65	7	4	2

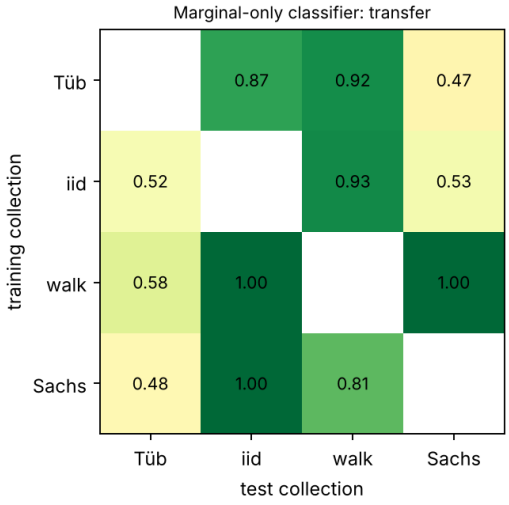


Figure 3. Marginal-only classifier trained on one collection (rows) and tested on another (columns), accuracy. The cue learned on Tübingen, Sachs or the chambers orients the chamber pairs; nothing learned on the chambers or Sachs transfers to Tübingen.

These first numbers leak. The chamber edges share their cause variable (`load_in` causes many sensors) and the 17 Sachs arcs reuse eleven proteins, so a held-out pair often had its variable's marginal in the training folds. Review of the first version identified the problem; experiment 52 removes it:

- i.i.d. chambers: training on the wind-tunnel recording and testing on the light-tunnel recording and conversely, 97.9% (93.6% without the unique-fraction feature);
- walk recordings: leaving one of the five recordings out, 86.9% (90.2%) (recordings of the same tunnel share variable names, so this is a weaker test than the wind-to-light transfer above, which is cross-device);
- Sachs: training on the arcs that involve neither protein of the test arc, 94.1% (all 17 arcs evaluated; same with or without the feature);
- Tübingen: leave one pair out, 57.8%.

The unique-fraction feature separates discrete actuators from continuous sensors and is a measurement artefact when sets are preprocessed differently (Tübingen is dequantised, so the feature is constant there); dropping it leaves the picture unchanged. The prediction held: above 80% without leakage.

Transfer across collections. A classifier trained on the 102 Tübingen pairs orients 87.2% of the i.i.d. chamber pairs and 91.8% of the walk pairs; one trained on the 17 Sachs pairs orients 100% of the i.i.d. chamber pairs (96.8% without the unique-fraction feature) and 81.1% of the walk pairs; one trained on the

Table 6. Tübingen: mean residual heteroscedasticity h (true direction) on the pairs a rule orients wrongly and rightly; p: one-sided Mann–Whitney test that h is larger on the wrong pairs. Twelve rules, no multiplicity correction.

Rule	wrong/right	h wrong	h right	p
IGCI	45/57	0.19	0.22	0.707
RECI	40/62	0.25	0.17	0.021
ANM-HSIC	40/62	0.32	0.13	0.000
Pairwise LiNGAM	40/62	0.22	0.19	0.245
PDM	61/41	0.22	0.18	0.021
LOCI	36/66	0.27	0.17	0.001
ANM, Gaussian process	42/53	0.29	0.10	0.000
Compression, first impl.	40/62	0.22	0.19	0.088
Compression, first impl., raw	36/66	0.22	0.19	0.095
TRA	29/53	0.26	0.17	0.004
Compression, faithful	34/68	0.23	0.19	0.075
AVICI	42/60	0.18	0.22	0.926
RESIT	37/60	0.34	0.13	0.000

chambers does not orient Tübingen (52% for the i.i.d.-trained and 58% for the walk-trained model); Sachs is oriented by the walk-trained model (100%, 88% without the unique-fraction feature) but not by the i.i.d.-trained one (53%; the Sachs set has 17 pairs). Figure 3 shows the matrix. The marginal asymmetry between an experimenter-set actuator and a sensor is a property shared by every device, and it is not present in Tübingen in a way that transfers: on Tübingen the marginal-only accuracy (57.8–59.0%) is within the sampling error of chance and of most dependence-based rules.

What this does and does not show. It does not show that RECI or IGCI use only the marginals: both read the scale of regression errors and the marginal density. It shows that the chamber collections can be oriented from the marginals alone, so a rule that is sensitive to marginal asymmetry will do well on them for reasons that have nothing to do with conditional structure, and that a rule that tests only the conditional model, such as an additive-noise test, has no reason to profit. That the chamber rankings (Figure 2) look like the marginal cue and the Tübingen ranking does not is consistent with this and is the first leg of the explanation.

9.2 Additive-noise rules work in one regime and reverse in another

Hypothesis. The additive-noise family loses where its assumption fails. For every Tübingen pair we measured the heteroscedasticity of the residuals of a cross-fitted additive-spline regression of the true effect on the true cause: $h = |\rho_S(|r|, x)|$ with ρ_S the Spearman correlation (six knots, ridge penalty one, two folds). Table 6 gives, for each rule, the mean h on the pairs it orients wrongly and rightly. The four noise-model rules show a large difference: ANM-HSIC 0.32 against 0.13, the Gaussian-process ANM 0.29 against 0.10, RESIT 0.34 against 0.13, LOCI 0.27 against 0.17 (one-sided Mann–Whitney, $p \leq 0.0014$ for each); also TRA ($p = 0.004$) and, weakly, RECI ($p = 0.021$) and PDM ($p = 0.021$); not IGCI ($p = 0.71$), pairwise LiNGAM ($p = 0.25$), AVICI ($p = 0.93$) or the compression rules ($p = 0.075$ to 0.095). Pooling the decisions of the four noise-model rules (396 decisions) in a logistic regression of correctness on the standardised h and $\log n$, one standard deviation of h multiplies the odds of a correct decision by 0.26 ($p < 10^{-18}$, optimistic because the four rules are scored on the same pairs); n has no effect ($p = 0.94$). We had predicted that RECI would not show the effect; it does, mildly.

Splitting Tübingen into thirds by h (bounds 0.094 and 0.268, 34 pairs each; TRA has 24, 31 and 27 pairs with a value) shows the regimes (Table 7, Figure 4). In the least heteroscedastic third ANM-HSIC, RESIT and the Gaussian-process ANM are right on 97%, 96% and 91% of the pairs, far above RECI (68%); in the most heteroscedastic third on 29%, 24% and 22%, below chance. Averaged over the three thirds, ANM-HSIC (0.61) and RESIT (0.62) equal the baselines and the Gaussian-process ANM is lower (0.55): a

Table 7. Tübingen accuracy by tercile of h (34 pairs per tercile; TRA 24, 31, 27).

Rule	low h	middle	high h
ANM-HSIC	0.97	0.56	0.29
RESIT	0.96	0.65	0.24
ANM, Gaussian process	0.91	0.53	0.22
TRA	0.83	0.68	0.44
LOCI	0.82	0.62	0.50
RECI	0.68	0.68	0.47
IGCI	0.59	0.44	0.65
Pairwise LiNGAM	0.65	0.53	0.65
Compression, faithful	0.74	0.74	0.53
Compression, first impl., raw	0.71	0.68	0.56
AVICI	0.53	0.56	0.68
PDM	0.56	0.35	0.29

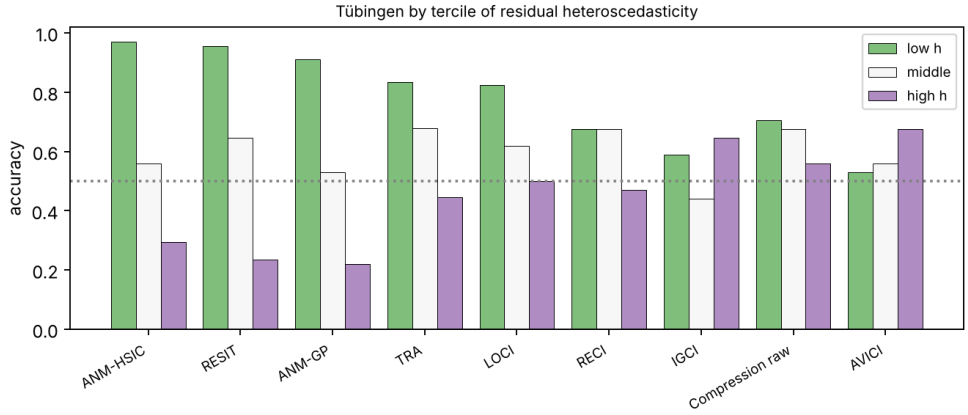


Figure 4. Tübingen accuracy by tercile of the residual heteroscedasticity h.

rule that is excellent in one regime and reversed in another has the accuracy of one that is mediocre in both. This reading is descriptive: h is computed in the true direction, so it is not a predictor; its bounds come from the same data; it is close to the dependence between the scale and the cause that the additive-noise tests probe, so part of the association is by construction.

9.3 Manipulation with a marginal control

An association across pairs cannot separate heteroscedasticity from whatever else differs between pairs, and making the noise scale depend on the cause also changes the marginal of the effect (a scale mixture). We therefore set the factor by hand (experiment 52). Pairs $e = f(c) + \sigma\varepsilon$, $f \in \{\tanh 2c, c^3/3, \sin 1.5c\}$, c uniform or Gaussian, $n = 500$, the same c, f and ε at every level, $a \in \{0, 0.5, 1, 1.5\}$, 100 pairs per level, a random half swapped. In the heteroscedastic arm $\sigma(c) \propto \exp(ac)$; in the control arm $\sigma_i \propto \exp(au_i)$ with $u_i \sim N(0, 1)$ independent of c, which gives the effect a similar heavy-tailed marginal but no dependence of the scale on the cause; the average noise variance is fixed in both. Figure 5 and Table 8: ANM-HSIC falls from 100% to 21% with cause-dependent noise and only to 89% with the independent scale, so the collapse is due to the dependence of the scale on the cause. IGCI and RECI rise in both arms (64 → 99% and 78 → 93% with cause-dependent noise; 64 → 99% and 78 → 98% with the independent scale): they profit from the heavy-tailed marginal of the effect, not from heteroscedasticity. An earlier unmatched run without the control showed the same rise and we had attributed it to heteroscedasticity; the control refutes

Table 8. Matched simulated pairs: accuracy against the strength a (100 pairs per level). Left: noise scale depends on the cause. Right: independent random scale with the same kind of marginal (control).

	scale depends on cause				independent scale			
Rule	0	0.5	1	1.5	0	0.5	1	1.5
ANM-HSIC	1.00	0.49	0.12	0.21	1.00	1.00	0.99	0.89
IGCI	0.64	0.72	0.91	0.99	0.64	0.73	0.95	0.99
RECI	0.78	0.77	0.87	0.93	0.78	0.84	0.96	0.98

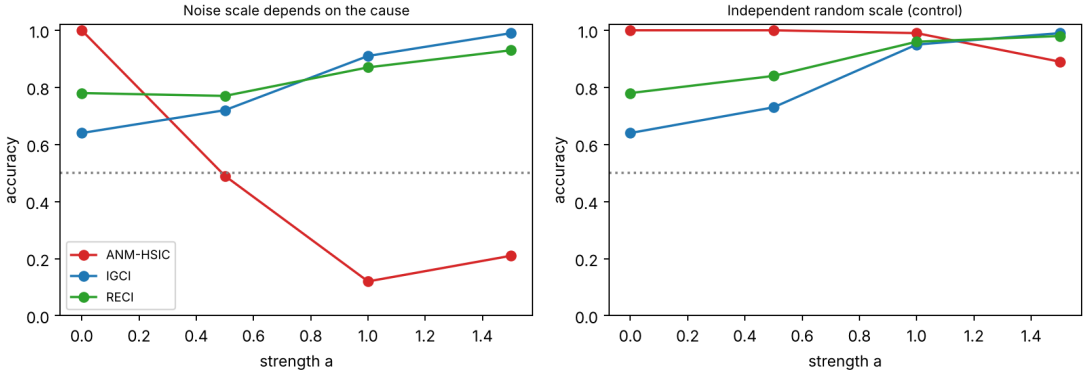


Figure 5. Accuracy against the strength a of the noise-scale variation. Left: scale depends on the cause. Right: independent scale (control).

that, and the correct reading ties this experiment to Section 9.1: IGCI and RECI are sensitive to marginal asymmetry.

9.4 A factorial map

To see the regimes together we crossed five factors (experiment 56): the mechanism ($\tanh 2c$, $c^3/3$, $\sin 1.5c$), the cause marginal (uniform, normal, bimodal, five-level discrete), the noise scale (homoscedastic, cause-dependent, independent random), measurement noise on both variables (0 or 0.3 sd), and $n \in \{200, 1000\}$: 144 cells of 25 pairs (3600 pairs), seven rules. Table 9 gives the accuracy by factor level and Table 10 the interaction of the noise scale with measurement noise.

The additive-noise rules are near-perfect only in the cell with homoscedastic noise and no measurement noise (ANM-HSIC 0.95, RESIT 0.95), fall to 0.69 and 0.74 with measurement noise alone, and to 0.06 and 0.17 with a cause-dependent scale; measurement noise partly masks the reversal (0.49, 0.54). IGCI and the faithful compression are the weakest rules when the cause is discrete (0.47 and 0.39); IGCI is more accurate with measurement noise than without (0.77 against 0.61); the faithful compression depends on n (0.62 at 200, 0.81 at 1000); AVICI fails on the cubic mechanism (0.19) and is at 0.49 for the uniform cause. RECI has the highest average accuracy in 12 of the 14 factor levels (the exceptions are the normal cause, where AVICI is at 0.89 against 0.72, and homoscedastic noise, where RESIT is at 0.84 against 0.82). Which factor breaks which rule is rule-specific.

9.5 An assumption-gated rule fails

If the regimes explain the Tübingen average, a rule that identifies the regime should beat both of its branches. We froze GATE on simulation before touching a real pair (experiment 51): $h_{\min} = \min(h(x \to y), h(y \to x))$, symmetric in the variables, threshold at the 95th percentile of $h_{\min}$ on 200 homoscedastic

Table 9. Factorial simulation (3600 pairs): accuracy of seven rules averaged over the other factors.

Factor	Level	IGCI	RECI	ANM-HSIC	LiNGAM	Compr.	RESIT	AVICI
mechanism	tanh	0.66	0.71	0.42	0.16	0.65	0.46	0.66
mechanism	cubic	0.87	1.00	0.70	0.54	0.75	0.75	0.19
mechanism	sin	0.55	0.95	0.73	0.81	0.74	0.75	0.85
cause marginal	uniform	0.94	0.95	0.52	0.34	0.84	0.53	0.49
cause marginal	normal	0.61	0.72	0.64	0.36	0.84	0.68	0.89
cause marginal	bimodal	0.75	0.89	0.63	0.59	0.79	0.68	0.36
cause marginal	discrete5	0.47	0.97	0.68	0.71	0.39	0.72	0.53
noise scale	homoscedastic	0.56	0.82	0.82	0.50	0.68	0.84	0.55
noise scale	cause	0.74	0.90	0.28	0.56	0.72	0.36	0.66
noise scale	independent scale	0.78	0.94	0.75	0.45	0.74	0.76	0.49
measurement noise	0	0.61	0.92	0.64	0.50	0.70	0.67	0.58
measurement noise	1	0.77	0.85	0.59	0.50	0.73	0.64	0.56
n	200	0.66	0.87	0.61	0.52	0.62	0.66	0.58
n	1000	0.73	0.90	0.62	0.48	0.81	0.65	0.55

Table 10. Factorial simulation: noise scale × measurement noise.

Noise scale	Meas. noise	IGCI	RECI	ANM-HSIC	LiNGAM	Compr.	RESIT	AVICI
homoscedastic	no	0.47	0.89	0.95	0.49	0.67	0.95	0.54
homoscedastic	yes	0.64	0.76	0.69	0.50	0.69	0.74	0.55
cause-dependent	no	0.68	0.91	0.06	0.55	0.71	0.17	0.68
cause-dependent	yes	0.81	0.89	0.49	0.56	0.73	0.54	0.65
independent scale	no	0.69	0.98	0.91	0.46	0.70	0.89	0.52
independent scale	yes	0.86	0.90	0.60	0.45	0.78	0.64	0.47

simulated pairs (0.058 at $n = 500$); GATE returns the ANM-HSIC direction when $h_{\min} \leq 0.058$ and the RECI direction otherwise. The pre-registered predictions were an accuracy of at least 0.70 on Tübingen and, on the chambers, within 0.05 of RECI. Both failed. On Tübingen GATE has 65.7% (77.1% with the official weights) against 60.8% (70.9%) for RECI; it routes 34% of the pairs to ANM-HSIC and the difference to RECI is not significant (exact McNemar 10 against 5, $p = 0.30$). On the i.i.d. chambers GATE has 75.5% against 97.9% (0 against 21 discordant pairs, $p < 10^{-6}$) and on the walks 77.0% against 94.3% ($p < 10^{-6}$), because it sends 88% and 62% of those pairs to ANM-HSIC, which is right on 72% and 63% of the whole collections (we did not store its accuracy on the routed subset). The threshold was set at $n = 500$, the chamber pairs have $n = 2000$ and the null spread of $|\rho|$ shrinks with n, so the routing is not comparable across sets; an n-aware threshold was not tried. The Sachs pairs were not evaluated (no stored ANM-HSIC decision under the same preprocessing). GATE is a negative result and is not offered as a method.

9.6 An attempt to predict real accuracy from the factorial map fails

The strongest test of an explanation is that it predicts. For every real pair we computed seven label-free diagnostics (minimum heteroscedasticity over the two directions, log fraction of unique values, minimum and maximum excess kurtosis, maximum absolute skewness, log absolute Spearman correlation, $\log n$), found its 100 nearest neighbours among the 3600 simulated pairs of Section 9.4, and predicted each rule's accuracy by the neighbours' mean correctness (experiment 57). The pre-registered criteria were a Spearman correlation of at least 0.5 between predicted and observed accuracy over the 49 (collection, rule) cells, a smaller RMSE than the constant prediction (each rule's overall simulation accuracy), and the rule predicted best being among the two best observed on at least four of the seven collections.

The first run was invalid as implemented: the unique-fraction diagnostic was constant in the simulation (computed on jittered arrays), so real pairs with tied values had standardised values around -3×10^9 and the neighbours were arbitrary (Spearman -0.01). Revision 2 computes simulated and real diagnostics identically (the simulator is deterministic) and reports how far the real pairs are from the simulation; it failed (Spearman 0.17, RMSE 0.30 against 0.26). Revision 3 replaces the observed decisions of the three new collections, which had been computed on jittered arrays, by the raw-array ones (Section 12); it is the version reported here. It fails as well: Spearman 0.006 (criterion 0.5), RMSE 0.31 against 0.24 for the constant prediction, while the criterion that the predicted best rule is among the two best observed is met on 4 of 7 collections (it is nearly degenerate: RECI is predicted best for every collection, and is among the two best observed on four of the seven; IGCI is observed best or second on five). The pair-level criterion, the AUC of the predicted correctness for the observed correctness, was also pre-registered and gives 0.66 for RESIT, 0.62 for ANM-HSIC, 0.55 for RECI and AVICI and 0.41 to 0.48 for IGCI, pairwise LiNGAM and compression, so the prediction carries some pair-level information for the two additive-noise rules and none for the others. The reason for the failure is visible in the support: the median distance from a real pair to its nearest simulated pair, in units of the simulation's standard deviations (a Euclidean metric on seven standardised diagnostics), is 2.3 for Tübingen, 2.8–3.4 for the chamber collections, 4.4–4.6 for the Bernoulli and changepoint collections, 8.9 for Sachs and 13.1 for Malus, against a 95th-percentile leave-one-out distance of 0.76 within the simulation; at most 5% of the pairs of any collection lie within the simulation's support. The factorial simulator illustrates mechanisms and cannot be used to predict accuracy on these real collections. (The simulated rules ran on jittered arrays; only the discrete-cause cells have ties, where IGCI's simulated accuracy would change with the tie convention.) The explanation of Sections 9.1–9.4 is therefore a set of demonstrated mechanisms, not a validated predictive model, and we do not claim that it accounts for the whole of the differences between the collections.

9.7 Summary of the evidence on mechanisms

Two mechanisms are directly supported. (1) The chamber and Sachs collections can be oriented from the marginals alone, validated across recordings and across proteins, and the same cue transfers across collections; IGCI and RECI are sensitive to marginal asymmetry and gain accuracy from a heavy-tailed effect marginal in simulation even when the conditional structure is unchanged. (2) Additive-noise rules are regime-specific: near-perfect when the noise scale does not depend on the cause and reversed when it does (manipulation with a control, factorial map, Tübingen terciles), and degraded by measurement noise. The Tübingen average of these rules equals the baselines because the collection mixes regimes. Two attempts to build on the account fail (GATE, predictive validation). What the evidence leaves unexplained is why the additive-noise rules lose on the chambers and how much of the differences between collections the two mechanisms jointly account for.

10 Graph benchmarks

The same question for graphs. We score skeletons only, against the 17-arc Sachs consensus (853 cells, log scale) and against the documented graphs of Causal Chambers recordings: two i.i.d. recordings (ten random subsamples of 4000 shuffled rows each), two further recordings with an effective sample size above 1000, and seventeen new wind-tunnel and light-tunnel recordings whose actuators are random walks and whose effective sample size, by the heuristic $n(1-\rho)/(1+\rho)$, is about three out of 4000 rows. For the walk recordings the first 4000 rows in time order are used. Table 11 gives precision/recall/F_1 for 22 method configurations: the classical rules (PC at two levels, GES, DirectLiNGAM, NOTEARS, DAGMA, PCMCI+), nine published packages at library defaults (experiment 45; AVICI, GOLEM, NOTEARS-MLP, GraN-DAG,

Table 11. Skeleton precision/recall/F_1 against the reference (strict), mean over data sets. Sachs: one data set; Chambers iid: 2×10 subsamples; ESS> 1000: 2 recordings; ESS< 20: 17 recordings. A number in parentheses is the number of data sets on which the method finished within 900 s when fewer than the full number. Experiments 36, 39, 43, 45.

Method	Sachs	Chambers iid	ESS > 1000	ESS < 20
PC α =0.05	1.00/0.41/0.58	0.40/0.38/0.37	0.48/0.28/0.35	0.36/0.39/0.37
PC α =0.01	1.00/0.35/0.52	0.39/0.33/0.34	0.51/0.27/0.36	0.36/0.36/0.36
GES-BIC	1.00/0.41/0.58	0.45/0.64/0.49	0.74/0.67/0.70	0.26/0.71/0.38
DirectLiNGAM	1.00/0.35/0.52	0.55/0.60/0.55	0.71/0.63/0.66	0.26/0.63/0.37
NOTEARS	1.00/0.18/0.30	0.41/0.21/0.27	0.33/0.24/0.26	0.26/0.45/0.33
DAGMA	1.00/0.29/0.45	0.44/0.33/0.36	0.44/0.41/0.39	0.25/0.52/0.34
PCMCI+	1.00/0.35/0.52	0.39/0.33/0.34	0.53/0.29/0.37	0.27/0.17/0.20
GRaSP	1.00/0.35/0.52	0.53/0.57/0.52	0.78/0.63/0.68	0.27/0.50/0.35
BOSS	1.00/0.35/0.52	0.57/0.61/0.55	0.77/0.57/0.65	0.29/0.54/0.37
BF-BIC + BOSS	0.90/0.53/0.67	0.43/0.63/0.49	0.57/0.61/0.58	0.30/0.68/0.42
NOTEARS-MLP	0.88/0.41/0.56	0.42/0.51/0.43	0.45/0.27/0.34 (1)	0.17/0.25/0.21 (1)
GOLEM	1.00/0.18/0.30	0.26/0.39/0.31	0.36/0.49/0.41	0.25/0.47/0.33
DAG-GNN	1.00/0.18/0.30	0.36/0.45/0.36	0.45/0.34/0.38	0.27/0.45/0.34 (16)
GraN-DAG	1.00/0.06/0.11	0.55/0.04/0.06	0.20/0.04/0.06	0.54/0.07/0.12
RESIT	0.39/0.53/0.45	0.33/0.80/0.43	0.37/0.74/0.49	0.23/0.99/0.37
CAM-UV	1.00/0.06/0.11	0.67/0.31/0.42	0.74/0.43/0.53	0.06/0.01/0.02 (15)
AVICI	0.88/0.41/0.56	0.48/0.36/0.37 (14)	0.54/0.17/0.26	0.36/0.23/0.28 (8)
PC-GCM α =0.05	0.89/0.47/0.62	0.38/0.33/0.34	0.54/0.34/0.41	0.35/0.54/0.42
PC-GCM α =0.01	1.00/0.47/0.64	0.39/0.30/0.32	0.47/0.26/0.33	0.34/0.46/0.39
BTAD	0.89/0.47/0.62	0.56/0.36/0.41	0.82/0.46/0.58	0.60/0.16/0.25
LRG	0.90/0.53/0.67	0.39/0.48/0.43	0.52/0.55/0.53	0.51/0.28/0.36
CERT-CD nonparametric	1.00/0.29/0.45	0.29/0.16/0.20	0.60/0.18/0.28	0.24/0.87/0.37

DAG-GNN, GRaSP, BOSS, RESIT, CAM-UV), BF-BIC with BOSS, and the methods we developed (PC with a nonparametric test, BTAD, LRG, sequential certificates, Appendix C).

The ranking depends on the family.

On the i.i.d. chamber recordings the best are DirectLiNGAM and BOSS ($F_1 = 0.55$) and GRaSP (0.52); on the two recordings with a large effective sample size GES (0.70), GRaSP (0.68) and DirectLiNGAM (0.66); on Sachs LRG and BF-BIC with BOSS (0.67), PC with the nonparametric test (0.62–0.64) and BTAD (0.62), against 0.58 for PC and GES. No method has a strict F_1 above 0.42 on the seventeen random-walk recordings, where most methods report tens of false edges. The neural methods are weaker: GraN-DAG finds almost no edges ($F_1 \leq 0.12$), GOLEM and DAG-GNN report 22 to 36 false edges per recording, NOTEARS with an MLP timed out on 16 of 17 walk recordings (16 variables) and AVICI on 9 of 17 and on 6 of 20 i.i.d. subsamples; their means in Table 11 are over the data sets on which they finished and are not comparable to the others.

Resolution and structure of the benchmarks.

One Sachs arc changes recall by 0.06; 16 of 22 configurations have a bootstrap interval that contains the best F_1 (Section 5). The chamber graphs are bipartite (actuators to sensors), so skeleton recovery is feature selection from actuators to measurements, and the actuators are recognisable by their marginals (Section 9.1). An interventional pilot on the light tunnel (a pilot, not pre-registered; `results/lt_interventions_check.json`) shows how little a graph benchmark with documented interventions leaves to a new method: a two-sample test of each sensor against the reference recording under each actuator intervention has precision 1.00 and recall 0.72 (28 of the 39 arcs among observed variables, which here are also the ancestral pairs; AUC 0.91 from the p-values), and the missed arcs, all from light

sources to their sensors, show no shift in the data (p between 0.06 and 0.86); a conditional version (residuals of a boosted regression on the estimated non-descendants) raises recall to 0.79 at a precision of 0.51.

What we do not conclude.
The reference networks are small, the Sachs consensus is contested, and several baselines (CAM, SCORE) were not run. The differences between graph methods are mostly within sampling error, and the effective sample size of the walk recordings limits what any method can show. We read the graph results as consistent with the pair results: no rule leads across families.

11 A reproduction audit

We audited one published table in detail, because it is the one for which the protocol is specified closely enough to repeat and for which the claims concern exactly the question of this paper. Stienstra [5] re-evaluate bivariate rules on the same 102 Tübingen pairs with the official weights, no tuning and a decision forced on every pair. They report 74.7% weighted accuracy for a parameter-free compression rule (68 of 102 pairs, 66.7% unweighted), 76.0% for the same rule on the 100 pairs on which SLOPE [13] is evaluated, 77.2% for SLOPE under a forced decision on those pairs (from the authors' stored output) and 82.4% as published (a decided-subset figure; 81.7% when the pairs answered by SLOPE's significance test are scored), and 70.7% for a faithful re-run of RECI; they trace the commonly quoted 77.5% for RECI to a mis-copied cell. Their point is that figures obtained with test-set model selection or with abstention gated by a significance test are not comparable with forced-decision figures.

What we reproduce.
Our RECI re-implementation gives 70.9% weighted (and 60.8% unweighted) on the same pairs, within 0.2 points of their 70.7%. Compression is the instructive case (Table 12). Our first implementation followed the published description (both variables quantised to 256 levels by linear min–max scaling, stable sort, first difference as `int8`, bzip2 level 9, normalised lengths, sign of the difference) but with the quantiser $\lfloor 256\,q \rfloor$ clipped to 255, and with the rows capped at 2000 and dequantised in the variant we ran at first; it gave 60.8% (dequantised) and 64.7% (raw rows), and 68.0% and 65.9% weighted. The text does not state how the integer levels are obtained from the scaled values. With $\mathrm{round}(255\,q)$, all rows (linearly thinned above 8000) and raw values the same code gives exactly 68 of 102 pairs and 74.66% weighted: the published figure. With $\lfloor 255\,q \rfloor$ it gives 71 pairs and 70.5%. A one-pair difference in the count moves the weighted accuracy by up to six points because the official weights are unequal. Capping the rows changes the faithful variant by at most four weighted points (71.7% at 500 rows, 74.4% at 2000, 74.7% at 8000, 75.3% without thinning); the conventions and caps are in `results/exp54_audit_conventions.json`. We adopted the faithful convention only after the reproduction check against the published figure, and we say so: it is chosen to match a published target, not to maximise accuracy on any data of ours.

What we do not reproduce or compare.
We did not run SLOPE, and so cannot confirm the 77.2% or 82.4% figures. For LOCI, our run of the authors' code gives 64.7% unweighted and 63.9% weighted (65.7% and 66.4% with 2000 rows instead of 1000); we did not locate a published Tübingen accuracy for LOCI that we could verify offline and make no comparison. The TRA preprint does not fix the constants we had to choose; our 64.6% on the 82 pairs it scores is for our constants and is not a test of its claims. SNOE, BF-BIC and AVICI make claims on graphs or simulated data that we did not attempt to reproduce.

Table 12. Reproduction of the compression baseline on the 102 Tübingen pairs (official weights). The published value is 68 of 102 pairs, 74.7% weighted.

Quantisation and rows	pairs correct	unweighted	weighted
$\lfloor 256q \rfloor$, all rows	67	0.657	0.688
$\lfloor 255q \rfloor$, all rows	71	0.696	0.705
$\mathrm{round}(255q)$, all rows	68	0.667	0.747
$\mathrm{round}(255q)$, 2000 rows	69	0.676	0.744
$\mathrm{round}(255q)$, no thinning	69	0.676	0.753
$\mathrm{round}(255q)$, 500 rows	71	0.696	0.717
first implementation (floor, 2000 rows)	62	0.608	0.680

What the audit shows for the argument.
(i) Two numbers that a reader would call "the compression baseline on Tübingen" differ by six weighted points because of an unreported rounding; a comparison between methods whose gaps are three to five points is therefore inside the variation produced by implementation conventions that no paper documents. (ii) The reproduced baseline is, on Tübingen, among the best rules in our panel (66.7%, 74.7% weighted), above RECI and LOCI but not significantly (exact McNemar $p = 0.39$ and 0.88), and it is at 38.3% on the i.i.d. chamber pairs, where RECI is at 97.9%. (iii) Its authors state that the rule's signal is partly marginal-geometric (a rank transform of the inputs, which flattens marginal shape, costs sixteen points in their experiments); this is the mechanism of Section 9.1.

12 Tie handling and implementation variants

The compression audit of Section 11 is one case of a general problem: choices that no paper reports and that a rule is sensitive to. Two more we measured.

Ties reverse IGCI.
The chamber arrays contain many tied values (discretised sensors, actuator levels). IGCI estimates the entropy of each scaled variable from the spacings between consecutive sorted values and prefers the direction in which the effect has the smaller entropy. Our implementation, like many, discards zero spacings; the alternative is to break ties with a tiny jitter, which turns the repeated values into spacings of order 10^{-6} and lowers the entropy of the tied variable. Table 13 gives the accuracy of IGCI under the two conditions on every collection (experiment 59, the same pairs): on the raw arrays 94% on the i.i.d. chamber pairs, 89% on the walk pairs, 100% on the Bernoulli and the changepoint pairs and 8% on Malus; with a jitter of one millionth of the range on the tied values 6%, 13%, 0%, 0% and 94%. The two conditions give the complementary accuracy on every collection that has ties (a against $1 - a$), so the decision of a rule that is "best on the chambers" or "best on Malus" is determined by a convention of one line of code. Tübingen, which is dequantised in both conditions and so has no ties, is unaffected (55.9%); on the raw Tübingen arrays, however, IGCI is at 63.8% on average over the protocol configurations of Section 8 against 52.4% when they are dequantised. RECI, pairwise LiNGAM and the cdt versions of RECI and CDS do not change; ANM-HSIC changes by up to ten points on one collection (changepoints, 0.57 raw, 0.47 jittered). TRA fails on the jittered arrays (its smoothing-spline fit is ill-posed) and works on the raw ones. We use the raw arrays for every rule (Section 3.2). A first pass of this study gave the new collections the jittered arrays, reported IGCI at 0% on the Bernoulli and the changepoint pairs and offered a conjecture about it; the conjecture was wrong, the cause was the jitter, and the stored raw results replace that analysis.

Table 13. Accuracy of rules under two tie conditions (raw arrays; 10^{-6} range jitter on tied values), same pairs (experiment 59). Tübingen is dequantised in both. cdt: `cdt` package defaults.

Rule	Ties	Tüb.	iid	walks	Sachs	Bern.	Chgpt.	Malus
IGCI (ours, uniform reference)	raw	0.56	0.94	0.89	0.65	1.00	1.00	0.08
IGCI (ours, uniform reference)	jitter	0.56	0.06	0.13	0.47	0.00	0.00	0.94
IGCI (cdt, Gaussian reference)	raw	0.42	0.94	0.84	0.47	1.00	1.00	0.08
IGCI (cdt, Gaussian reference)	jitter	0.42	0.06	0.12	0.53	0.00	0.00	0.94
ANM-HSIC	raw	0.57	0.73	0.62	0.59	0.31	0.57	0.31
ANM-HSIC	jitter	0.57	0.72	0.62	0.59	0.31	0.47	0.33
RECI	raw	0.61	0.98	0.94	0.53	0.25	1.00	0.56
RECI	jitter	0.61	0.98	0.94	0.53	0.25	1.00	0.56
Pairwise LiNGAM	raw	0.61	0.78	0.71	0.53	0.88	0.43	0.40
Pairwise LiNGAM	jitter	0.61	0.78	0.71	0.53	0.88	0.43	0.40

Reference implementations.

Running the `cdt` implementations (experiment 55) gives a check of ours. Our IGCI makes exactly the decisions of cdt's IGCI with a uniform reference measure on every pair of every collection (100% agreement), whereas cdt's default reference is Gaussian (standardised variables): on Tübingen the two references give 55.9% and 42.2% (80% of the decisions agree) and on the raw Sachs pairs 64.7% and 47.1%; on the i.i.d. chamber pairs they agree (94% in both, raw) and on the walks they differ little (89% and 84%). The IGCI reference measure is thus a free parameter with a 14-point effect on Tübingen. Our RECI makes the same decision as cdt's RECI on 98.9% of the i.i.d. chamber pairs, 97.5% of the walk pairs and all pairs of the other chamber collections, Sachs and the new collections, but on 85.3% of the Tübingen pairs (60.8% against 63.7% accuracy on Tübingen): the two differ in the regression used for the error. cdt's CDS reaches 61.8% on Tübingen, 90.4% on the i.i.d. chambers and 79.5% on the walks; cdt's ANM (Gaussian-process regression with an HSIC test) and BivariateFit, run on the jittered arrays of experiment 55, are at 51.5% and 48.0% on Tübingen and at 46% and 5% on the i.i.d. chambers. We did not obtain the authors' code for these rules; the cdt versions are themselves re-implementations.

Seed variability.

The HSIC-based rule has a random component: the same ANM-HSIC on the same Tübingen pairs gives 60.8% with the seed of the first experiments and 56.9% with the seed of experiment 59 (2000 rows, dequantised). Differences of this size are within what separates published rules.

13 Discussion and recommendations

What the evidence supports.

On the real collections we examined, the rules that test a model of the conditional distribution do not beat rules that read marginal asymmetry or error scale, and no rule leads across collections. This is not because the dependence-based rules are poor: ANM-HSIC, RESIT and the Gaussian-process ANM are right on 91 to 97% of the least heteroscedastic third of the Tübingen pairs, and ANM-HSIC is perfect in simulation when the assumption holds. They are reversed when the noise scale depends on the cause, degraded by measurement noise, and unable to use the marginal cue that orients the chamber collections. Averaged over a collection that mixes regimes they equal the baselines. The baselines are not robust either: IGCI is at 8% on the Malus pairs and, depending on whether ties are broken with a jitter, at 94% or 6% on the i.i.d. chambers; RECI is at 25% on the Bernoulli pairs; compression is at 38% on the i.i.d. chambers.

What it does not support.
It does not show that any published method misreports; we audited one table and reproduced it. It does not show that dependence-based rules cannot be improved: the failed GATE and the failed predictive validation show that our account is incomplete, and that a regime switch on a single diagnostic is not enough. It does not show that the same pattern holds on other collections of pairs; the collections of Table 1 are small and, outside Tübingen, concern one or two devices.

Implications for evaluation.
1. *Test paired and state the power.* Compare rules with an exact paired test on the same pairs and report the detectable difference. With 102 pairs it is about 19 points, 27 after correcting for 78 comparisons; a gain of two to five points is not resolvable.
2. *Report the coverage curve and both weightings.* A single decided-only number, a weighted number alone, or an unweighted number alone each select one point of a multiverse that moves a rule by 11 to 26 points and its rank by up to ten places.
3. *Fix subsets, preprocessing and settings before the data are seen, and publish code that applies them,* including conventions the text does not fix: one rounding rule moved a weighted headline by six points, the reference measure of IGCI moved its accuracy by fourteen points on Tübingen, and the treatment of tied values turned 94% into 6% on the i.i.d. chambers.
4. *Report several collections, and test the marginal-only classifier.* A rule evaluated on one collection tells little about another (Kendall τ between the Tübingen and chamber rankings is negative). A marginal-only classifier is cheap to run on a new collection and indicates how much of the benchmark can be solved without dependence structure.
5. *Report the effective sample size and the number of distinct edges.* Seventeen of nineteen new chamber recordings have an effective sample size of about three; three of our new pair collections consist of four to eight distinct edges.
6. *For learned methods, show real-data accuracy next to the simulated test*, and describe how far the real diagnostics lie from the simulator's support (Section 9.6).

Why not simply propose a better rule?
We tried, and report the attempts (Appendix C): sequential certificates, a block-replicated additive method, a learned graph classifier, a learned pair classifier, an invariance method with time blocks, a regime-gated rule. None beats the baselines on the real collections. We consider the negative results informative because each was fixed before it was run and the failures are explained, in part, by the mechanisms of Section 9: a conditional-structure rule is up against a marginal cue and a mixture of regimes.

14 Limitations

– *Size.* Tübingen has 102 pairs and the Sachs data 17 arcs; the other pair collections have four to 47 distinct edges. All inference is as imprecise as Section 5 says.
– *Implementations.* IGCI, RECI, ANM-HSIC and pairwise LiNGAM are short re-implementations that we did not check against the authors' code on the Tübingen pairs (RECI agrees with a published faithful re-run within 0.2 weighted points); statements about them are about our implementations. Twelve rules were run by us, some at library defaults and two re-implemented from preprints (compression, now faithful; TRA with our constants). Tuning could change a ranking. We did not run SLOPE, CAM, SCORE or PNL.

– *The multiverse* is finite, its choices are ours, its accuracies are computed on overlapping pairs and are not independent, and several scores are not confidences. The detectable difference of 0.19 applies to the observed discordance rate and is approximate.
– *The heteroscedasticity measure* is computed in the true direction and is close to what the additive-noise tests probe; the tercile analysis is post hoc. The manipulation and the factorial map are simulations from one family of mechanisms.
– *The factorial simulator* does not cover the real collections (Section 9.6).
– *The marginal-only classifier* is small, with five features, and its chamber and Sachs accuracies rest on collections with few distinct causes; the Tübingen result is within the sampling error of chance.
– *Unexplained.* Why additive-noise rules lose on the chambers, and why IGCI fails on Malus (where the cause has about 730 levels and the dependence is non-monotone) are not diagnosed beyond the observation.
– *Process.* Analyses designed after the data are labelled as such; several errors found on internal review (leakage in the marginal classifier, a confounded manipulation, universes with too few pairs) are corrected in the text and described where they occurred.

Data and code availability.
All code, stored per-pair decisions, scripts and pre-registrations are in the repository `vinhqdang/SPRINT-CD`. The Tübingen pairs, the Sachs data and the Causal Chambers data are public; the code of LOCI is fetched from its authors' repository at run time.

Declarations

Funding information: [to be completed by the author].
Author contributions: single author.
Conflict of interest: [to be completed by the author].
Data availability statement: the Tübingen pairs, the Sachs data and the Causal Chambers data are public; all code, stored per-pair decisions, scripts and pre-registrations are in the repository `vinhqdang/SPRINT-CD` (a version-archived copy with a DOI will be deposited before publication).

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A Pre-registration log

Each row is an experiment script (`experiments/expNN_*.py`) with the short hash of the commit that first added it, before it was run, and the date. The first line of each script's docstring is quoted. Scripts were revised after a first run only where stated in their docstrings (experiments 47, 49, 57; each revision is a separate commit and the first version is retrievable). Experiments 1–16 belong to the earlier theory paper on sequential certificates and are not used here.

#	Commit	Date	Purpose
17	`98de084`	2026-10-23	Experiment 17: scaling with graph size and ablation of CERT-CD.
18	`0318781`	2026-10-23	Experiment 18: held-out simulated pair families, sample-size sweep, baselines.
19	`0318781`	2026-10-23	Experiment 19: sensitivity of the Tuebingen result to the random seed.
20	`c5cd9b0`	2026-10-23	Experiment 20: real physical data with a known causal graph (Causal Chambers).
21	`269ae7a`	2026-10-23	Experiment 21: direction of the known edges of the Causal Chambers.
22	`aa99d05`	2026-10-23	Experiment 22: nonparametric sequential adjacency certificate vs the linear primitive.
23	`aa99d05`	2026-10-23	Experiment 23: nonparametric adjacency certificate on the Causal Chambers.
24	`9f32aaa`	2026-10-23	Experiment 24: Causal Chambers with conditioning-set order k = 3.
25	`7045b1b`	2026-10-23	Experiment 25: the nonparametric certificate on the exp13 and exp17 DAG data.
26	`b59a1c6`	2026-10-23	Experiment 26: PC with the same nonparametric test, under monitoring.
27	`ba2f7ac`	2026-10-23	Experiment 27: untouched Causal Chamber recordings, final algorithm frozen.
28	`06bffaf`	2026-10-23	Experiment 28: abstaining comparison of the pair rules at matched coverage.
29	`f85302f`	2026-10-23	Experiment 29: BTAD against PC, GES, DirectLiNGAM and PC with an iid nonparametric test
30	`ced1b11`	2026-10-23	Experiment 30: BTAD on the untouched Causal Chamber recordings, time-ordered.
31	`9dddc37`	2026-10-24	Fit the LRG classifier on exp31 features; choose the threshold on validation; report held-out simulation metrics.
31	`d682048`	2026-10-24	Experiment 31: train the learned replicated-gain skeleton model (LRG) on simulated data.
32	`9dddc37`	2026-10-24	Experiment 32: the frozen LRG model on the untouched, time-ordered Causal Chamber recordings.

#	Commit	Date	Purpose
33	b91cbf1	2026-10-24	Experiment 33: frozen LRG and BTAD on the two Causal Chamber recordings with i.i.d. random actuators.
34	985335e	2026-10-24	Experiment 34: train the learned pair-direction model (PDM) on simulated pairs.
34	ddb1d8f	2026-10-24	Fit the pair-direction classifier on exp34 features; report held-out simulated accuracy.
35	ddb1d8f	2026-10-24	Experiment 35: the frozen pair-direction model (PDM) on real and held-out pair sets.
36	5e7f00b	2026-10-24	Experiment 36: Sachs data and NOTEARS, added to the graph benchmark.
37	a83a699	2026-10-24	Experiment 37: PC with the nonparametric test on the two i.i.d. chamber recordings (exp20 subsamples).
38	c8e01d1	2026-10-24	Experiment 38: LOCI and ANM added to the pair benchmark, and the Sachs pairs.
39	6387fdc	2026-10-24	Experiment 39: DAGMA and PCMCI+ added to the graph benchmark.
40	ba11744	2026-10-24	Experiment 40: LOCI on the Tuebingen pairs with up to 2000 rows per pair.
41	c94369a	2026-10-24	Experiment 41 (negative result): block-ICP pilot on five wind-tunnel walk recordings (development only).
42	bbdeb3a	2026-10-24	Experiment 42: 2025-2026 pair rules on the same pair sets as exp38.
43	bbdeb3a	2026-10-24	Experiment 43: BF-BIC with BOSS (arXiv 2510.04276) on the graph benchmark.
44	cdf776c	2026-10-24	Experiment 44: SNOE-style edge orientation (arXiv 2506.05590, see sprint_cd/snoe_lite.py).
45	37f8d52	2026-10-25	Experiment 45: published methods with public code on the graph benchmark.
46	101841b	2026-10-25	Experiment 46: published pair rules with public code (AVICI pretrained, RESIT) on the exp38 pair sets.
47	b805f02	2026-10-25	Experiment 47 (revision 4: the discordance rate and McNemar counts exclude missing values and exact ties, the Bonferroni
48	211fb35	2026-10-25	Experiment 48: resolution of the Sachs graph benchmark.
49	02dfe48	2026-10-25	Experiment 49: why do direction rules that exploit dependence lose to RECI/IGCI on real pairs?
50	c17d1ad	2026-10-25	Experiment 50: does heteroscedastic noise cause the loss of the noise-model rules? (simulation, dose-response)
51	31dadda	2026-10-25	Experiment 51: an assumption-gated direction rule (GATE), frozen on simulation, tested once on real pairs.
52	4282166	2026-10-25	Experiment 52: controls for experiments 49-50 (leakage-free marginal classifier, matched dose-response with marginal con
53	37fc958	2026-10-25	Experiment 53: three further real pair collections from the Causal Chambers, and the diagnostics of exp49.
54	37d6d13	2026-10-25	Experiment 54: sorted-conditional compression with the quantisation convention that reproduces its published figure.
55	8f63bb7	2026-10-25	Experiment 55: reference implementations from the cdt package, and LOCI/ANM-GP/PDM on the three new collections.
56	04de9ce	2026-10-25	Experiment 56: a factorial simulation of the regimes found in exp49-52, with label-free diagnostics.
57	241d7b0	2026-10-25	Experiment 57 (revision 3: the observed decisions of the three new collections are the raw-array ones of exp59 (IGCI, RE
58	5c1c168	2026-10-25	Experiment 58: pair rules on every untouched Causal Chambers recording (not only the first of each type).
59	8b388ce	2026-10-25	Experiment 59: tie handling as a free parameter, and raw-array versions for the three new collections.
60	b592bf7	2026-10-25	Experiment 60: TRA on raw arrays for the all-recordings walk pairs (exp58) and the three new collections (exp53).

B Implementation details

Preprocessing.
Tübingen: at most 2000 rows drawn with a seed that depends on the pair identifier; each variable with tied values dequantised by adding uniform noise in $\pm\frac{1}{2}$ of the smallest positive gap between distinct values (experiment 14). Chamber and Sachs collections: raw values (Sachs on the log scale with a floor at 10^{-3}), at most 2000 random rows; walk recordings use random rows of the first 4000 rows for pairs and the first 4000 (graphs) or 2000 (45) rows in time order for graphs. For the three new collections, rules that need distinct values receive a jitter of 10^{-6} times the range on tied values.

Scores.
All rules return a signed score, positive for $x \to y$. IGCI, RECI, ANM-HSIC and pairwise LiNGAM return a direction label and a magnitude that we convert to a sign. A score of exactly zero counts as half correct (this affected 23 pairs of the Gaussian-process ANM). A rule that returns no value on a pair (TRA's spline failure) is excluded from that pair and the denominator is reduced.

Compression.
Section 11. In both variants the score is $s = \mathrm{sc}(q_y, q_x)/\max(\mathrm{dc}(q_x), 1) - \mathrm{sc}(q_x, q_y)/\max(\mathrm{dc}(q_y), 1)$ with $\mathrm{sc}(\mathrm{key}, \mathrm{val}) = |\mathrm{bz2}(\Delta\,\mathrm{val}[\mathrm{argsort}(\mathrm{key})])|$ and $\mathrm{dc}(q) = |\mathrm{bz2}(\Delta q)|$, a stable sort, differences cast to `int8`, bzip2 level 9; $s > 0$ means $x \to y$.

TRA.
Cross-fitted smoothing-spline residuals (5 folds), a rank-uniform point cloud, the total length of the zero-dimensional persistence (minimum spanning tree) in the window $\alpha = \kappa n^{-2/3}$, $\beta = c_\beta \alpha$ with $\kappa = 1$, $c_\beta = 4$, at most 1000 rows; the direction is the sign of the difference of the two residual functionals. The conservative variant TRA-C ($B = 100$ Gaussian-copula bootstrap draws) took 27 s per pair at $n = 800$ and was not run.

AVICI.
The pretrained model `scm-v0`; edge probability matrix of the standardised data (≤ 2000 rows for graphs, ≤ 1000 for pairs); graphs: skeleton from probability ≥ 0.5; pairs: $P(x \to y) - P(y \to x)$. Run in a virtual environment with `jax 0.4.30`, `flax 0.8.5`, `dm-haiku 0.0.12`, `optax 0.2.2`, `numpy<2`.

Graph packages.
GOLEM (gCastle, 10^4 iterations), NOTEARS-MLP and GraN-DAG and DAG-GNN (gCastle defaults), GRaSP and BOSS (`causal-learn`, BIC), RESIT (random forest depth 4, $\alpha = 0.01$) and CAM-UV ($\alpha = 0.01$, three explanatory values; `lingam`); a run over 900 s is a timeout.

Marginal-only classifier.
Features as in Section 9.1; scikit-learn `LogisticRegression` with $C = 1$ on standardised features; both orientations of every training pair; 50 repetitions of five-fold cross-validation for the first protocol, deterministic splits for the others.

Heteroscedasticity.
Cubic B-splines with six knots on the standardised cause, ridge penalty one, two-fold cross-fitting, residual scale correlation $|\rho_S(|r|, x)|$.

Factorial simulation.
Seeds are `default_rng([20263200, mechanism, cause, scale, measurement, n, k])`; the scale of the noise is normalised so that the mean noise variance is $(0.3\,\mathrm{sd}\,f(c))^2$; measurement noise is Gaussian with 0.3 times the sample sd of each variable.

Software.
Python 3.11, `numpy 1.26`, `scipy 1.13`, `scikit-learn 1.9`, `statsmodels 0.15`, `causal-learn 0.1.4.8`, `lingam 1.13`, `gcastle`, `causalchamber 0.2.8`; four CPU cores, no GPU.

C Negative results of the algorithm search

Each attempt was fixed before its run. Numbers are strict skeleton F_1 or forced accuracy against the reference.

Attempt	Idea	Outcome on real data
CERT-CD (sequential certificates)	assert edges only by rejecting; anytime-valid	False-edge control holds in simulation (2.7% against 27–91%), not on the chambers
BTAD	block-replicated additive gains	highest precision (0.60) but recall 0.16; F_1 0.25–0.28 on walk recordings
LRG	gradient boosting on block-gain features, trained on simulations	F_1 0.36 (chambers), 0.67 (Sachs); within-recording AUC 0.55
PDM	pair classifier trained on 40000 simulated pairs	95% on simulated pairs, 40.2% on Tübingen
Block-ICP	invariant causal prediction with time blocks	empty graphs, $F_1 \leq 0.07$ on development recordings
Compression, TRA, BF-BIC (exp 42–43, 54)	re-implemented 2025–2026 methods	compression: 60.8–64.7% with our first implementation, 66.7% (74.7% weighted) faithful, 38.3% on i.i.d. chambers; TRA 64.6% (86.2% on i.i.d. chambers); BF-BIC with BOSS: Sachs 0.67, walks 0.42
Interventional route (pilot)	shift of each sensor under each intervened actuator (light tunnel)	KS: precision 1.00, recall 0.72 (28 of 39 arcs), AUC 0.93; the missed arcs show no shift; conditional version recall 0.79 at precision 0.51
Published packages (exp 45–46)	AVICI, GOLEM, NOTEARS-MLP, GraN-DAG, DAG-GNN, GRaSP, BOSS, RESIT, CAM-UV	best: BOSS 0.55 i.i.d. chambers; GRaSP 0.68, BOSS 0.65 large-ESS; Sachs ≤ 0.56; walks ≤ 0.37; many neural timeouts
GATE (exp 51)	ANM-HSIC when a symmetric heteroscedasticity diagnostic is low, else RECI	Tübingen 65.7% against 60.8% ($p = 0.30$); chambers 75.5%, 77.0% against 97.9%, 94.3%
Predictive validation (exp 57)	nearest simulated neighbours predict each rule's accuracy	Spearman 0.17 (needed 0.5); RMSE 0.30 against 0.26 for the constant; real pairs are 2–13 sd from the simulation

D Additional tables

Table 19. Factorial simulation, accuracy per cell (25 pairs each; cause marginal, noise scale, measurement noise, n; rules in the order IGCI, RECI, ANM-HSIC, LiNGAM, compression (faithful), RESIT, AVICI).

Mech.	Cause	Scale	Meas.	n	IGCI	RECI	ANM-H	LiNGAM	Cmpr.	RESIT	AVICI
tanh	uniform	homo	n	200	0.88	1.00	1.00	0.00	0.68	0.92	0.56
tanh	uniform	homo	n	1000	1.00	1.00	1.00	0.00	1.00	1.00	0.20
tanh	uniform	homo	y	200	0.72	0.72	0.16	0.04	0.64	0.36	0.44
tanh	uniform	homo	y	1000	0.80	0.68	0.00	0.00	0.68	0.00	0.44
tanh	uniform	cause	n	200	1.00	1.00	0.00	0.04	0.84	0.00	1.00
tanh	uniform	cause	n	1000	1.00	1.00	0.00	0.00	1.00	0.00	1.00
tanh	uniform	cause	y	200	0.60	0.72	0.04	0.16	0.72	0.00	0.84
tanh	uniform	cause	y	1000	1.00	0.92	0.00	0.04	0.72	0.00	1.00
tanh	uniform	indep	n	200	1.00	1.00	0.80	0.00	0.88	0.80	0.32
tanh	uniform	indep	n	1000	1.00	1.00	0.96	0.12	1.00	1.00	0.52
tanh	uniform	indep	y	200	0.84	0.76	0.12	0.00	0.52	0.16	0.32
tanh	uniform	indep	y	1000	1.00	1.00	0.08	0.00	1.00	0.00	0.44
tanh	normal	homo	n	200	0.12	0.12	1.00	0.00	0.68	1.00	1.00
tanh	normal	homo	n	1000	0.00	0.04	1.00	0.00	1.00	1.00	1.00
tanh	normal	homo	y	200	0.12	0.00	0.80	0.00	0.36	0.84	1.00

Mech.	Cause	Scale	Meas.	n	IGCI	RECI	ANM-H	LiNGAM	Cmpr.	RESIT	AVICI
tanh	normal	homo	y	1000	0.00	0.00	0.96	0.00	0.48	1.00	1.00
tanh	normal	cause	n	200	0.80	0.36	0.32	0.16	0.88	0.20	1.00
tanh	normal	cause	n	1000	1.00	0.72	0.32	0.16	1.00	0.32	1.00
tanh	normal	cause	y	200	0.48	0.28	0.52	0.16	0.48	0.56	1.00
tanh	normal	cause	y	1000	0.48	0.40	0.64	0.04	0.76	0.68	1.00
tanh	normal	indep	n	200	0.92	0.64	1.00	0.04	0.84	1.00	1.00
tanh	normal	indep	n	1000	0.96	0.92	1.00	0.04	1.00	1.00	1.00
tanh	normal	indep	y	200	0.20	0.12	0.56	0.04	0.52	0.56	1.00
tanh	normal	indep	y	1000	0.64	0.48	0.60	0.00	0.72	0.84	1.00
tanh	bimodal	homo	n	200	0.72	0.56	0.44	0.40	0.68	0.44	0.56
tanh	bimodal	homo	n	1000	0.68	0.60	0.56	0.12	0.76	0.60	0.56
tanh	bimodal	homo	y	200	0.52	0.56	0.48	0.44	0.56	0.60	0.48
tanh	bimodal	homo	y	1000	0.40	0.36	0.40	0.16	0.36	0.48	0.32
tanh	bimodal	cause	n	200	1.00	0.20	0.00	0.64	0.92	0.00	0.84
tanh	bimodal	cause	n	1000	1.00	0.48	0.00	0.92	1.00	0.00	0.72
tanh	bimodal	cause	y	200	0.84	0.72	0.04	0.84	0.44	0.08	0.84
tanh	bimodal	cause	y	1000	0.96	0.88	0.04	0.72	0.84	0.00	0.76
tanh	bimodal	indep	n	200	1.00	1.00	0.28	0.04	0.52	0.20	0.16
tanh	bimodal	indep	n	1000	1.00	1.00	0.28	0.00	1.00	0.68	0.04
tanh	bimodal	indep	y	200	0.92	0.80	0.32	0.16	0.56	0.52	0.40
tanh	bimodal	indep	y	1000	1.00	1.00	0.16	0.00	1.00	0.20	0.08
tanh	discrete5	homo	n	200	0.00	1.00	1.00	0.00	0.00	1.00	0.32
tanh	discrete5	homo	n	1000	0.00	1.00	1.00	0.00	0.00	1.00	0.08
tanh	discrete5	homo	y	200	0.48	0.52	0.20	0.00	0.60	0.28	0.64
tanh	discrete5	homo	y	1000	0.76	0.76	0.08	0.00	0.52	0.08	0.72
tanh	discrete5	cause	n	200	0.00	1.00	0.00	0.48	0.00	0.16	0.84
tanh	discrete5	cause	n	1000	0.00	1.00	0.04	0.40	0.00	0.08	1.00
tanh	discrete5	cause	y	200	0.88	0.92	0.12	0.40	0.64	0.12	0.92
tanh	discrete5	cause	y	1000	1.00	1.00	0.04	0.44	0.88	0.00	1.00
tanh	discrete5	indep	n	200	0.00	1.00	0.88	0.08	0.00	1.00	0.36
tanh	discrete5	indep	n	1000	0.00	1.00	1.00	0.04	0.00	1.00	0.24
tanh	discrete5	indep	y	200	0.92	0.92	0.00	0.16	0.76	0.36	0.56
tanh	discrete5	indep	y	1000	1.00	1.00	0.00	0.00	0.96	0.04	0.24
cubic	uniform	homo	n	200	1.00	1.00	1.00	0.96	0.84	1.00	0.00
cubic	uniform	homo	n	1000	1.00	1.00	1.00	1.00	1.00	1.00	0.00
cubic	uniform	homo	y	200	0.96	1.00	1.00	0.96	0.84	1.00	0.00
cubic	uniform	homo	y	1000	1.00	1.00	1.00	1.00	1.00	1.00	0.00
cubic	uniform	cause	n	200	1.00	1.00	0.00	0.40	0.88	0.08	0.00
cubic	uniform	cause	n	1000	1.00	1.00	0.00	0.32	1.00	0.12	0.00
cubic	uniform	cause	y	200	1.00	1.00	0.60	0.40	0.48	0.80	0.00
cubic	uniform	cause	y	1000	1.00	1.00	0.52	0.44	1.00	0.80	0.00
cubic	uniform	indep	n	200	1.00	1.00	1.00	0.84	0.92	1.00	0.24
cubic	uniform	indep	n	1000	1.00	1.00	1.00	0.68	1.00	1.00	0.00
cubic	uniform	indep	y	200	1.00	1.00	0.96	0.76	0.56	1.00	0.00
cubic	uniform	indep	y	1000	1.00	1.00	0.96	0.80	1.00	1.00	0.00
cubic	normal	homo	n	200	1.00	1.00	0.96	0.08	1.00	1.00	0.84
cubic	normal	homo	n	1000	1.00	1.00	1.00	0.04	1.00	0.96	0.92
cubic	normal	homo	y	200	1.00	1.00	0.40	0.04	0.64	0.88	0.80
cubic	normal	homo	y	1000	1.00	1.00	0.24	0.00	1.00	0.52	0.48
cubic	normal	cause	n	200	1.00	1.00	0.00	0.08	1.00	0.16	0.80
cubic	normal	cause	n	1000	1.00	1.00	0.00	0.00	1.00	0.00	0.48
cubic	normal	cause	y	200	1.00	1.00	0.04	0.00	0.64	0.56	0.52
cubic	normal	cause	y	1000	1.00	1.00	0.00	0.04	1.00	0.04	0.16
cubic	normal	indep	n	200	1.00	1.00	0.84	0.16	0.96	0.68	0.92
cubic	normal	indep	n	1000	1.00	1.00	1.00	0.32	1.00	0.28	0.96
cubic	normal	indep	y	200	1.00	1.00	0.24	0.00	0.56	0.64	0.64
cubic	normal	indep	y	1000	1.00	1.00	0.16	0.00	1.00	0.16	0.56
cubic	bimodal	homo	n	200	0.96	1.00	1.00	0.80	0.92	1.00	0.00
cubic	bimodal	homo	n	1000	1.00	1.00	1.00	0.44	1.00	1.00	0.00
cubic	bimodal	homo	y	200	1.00	1.00	1.00	0.64	0.48	1.00	0.00
cubic	bimodal	homo	y	1000	1.00	1.00	1.00	0.52	1.00	1.00	0.00
cubic	bimodal	cause	n	200	1.00	1.00	0.00	0.12	0.88	0.00	0.00
cubic	bimodal	cause	n	1000	1.00	1.00	0.00	0.00	1.00	0.00	0.00
cubic	bimodal	cause	y	200	0.96	1.00	0.88	0.44	0.24	0.92	0.00
cubic	bimodal	cause	y	1000	1.00	1.00	0.96	0.08	1.00	1.00	0.00
cubic	bimodal	indep	n	200	0.96	1.00	0.96	0.64	0.88	0.92	0.00
cubic	bimodal	indep	n	1000	1.00	1.00	1.00	0.28	1.00	1.00	0.00
cubic	bimodal	indep	y	200	1.00	1.00	1.00	0.64	0.44	0.96	0.00
cubic	bimodal	indep	y	1000	1.00	1.00	1.00	0.28	1.00	1.00	0.00
cubic	discrete5	homo	n	200	0.00	1.00	1.00	1.00	0.00	1.00	0.00
cubic	discrete5	homo	n	1000	0.00	1.00	1.00	1.00	0.00	1.00	0.00
cubic	discrete5	homo	y	200	0.96	0.96	1.00	1.00	0.72	1.00	0.00
cubic	discrete5	homo	y	1000	1.00	1.00	1.00	1.00	0.88	1.00	0.00
cubic	discrete5	cause	n	200	0.00	1.00	0.04	1.00	0.00	0.20	0.12
cubic	discrete5	cause	n	1000	0.00	1.00	0.04	1.00	0.00	0.60	0.00
cubic	discrete5	cause	y	200	1.00	1.00	0.80	1.00	0.68	0.88	0.00
cubic	discrete5	cause	y	1000	1.00	1.00	1.00	1.00	0.92	1.00	0.00
cubic	discrete5	indep	n	200	0.00	1.00	1.00	0.96	0.00	1.00	0.24
cubic	discrete5	indep	n	1000	0.00	1.00	1.00	0.88	0.00	1.00	0.20
cubic	discrete5	indep	y	200	0.92	1.00	1.00	0.96	0.56	1.00	0.04

Mech.	Cause	Scale	Meas.	n	IGCI	RECI	ANM-H	LiNGAM	Cmpr.	RESIT	AVICI
cubic	discrete5	indep	y	1000	1.00	1.00	1.00	0.96	1.00	1.00	0.00
sin	uniform	homo	n	200	0.96	1.00	0.96	0.04	0.84	0.84	0.88
sin	uniform	homo	n	1000	1.00	1.00	1.00	0.00	1.00	1.00	1.00
sin	uniform	homo	y	200	0.60	0.80	0.48	0.12	0.84	0.52	0.64
sin	uniform	homo	y	1000	0.80	0.88	0.40	0.00	0.64	0.28	0.84
sin	uniform	cause	n	200	0.96	1.00	0.00	0.76	1.00	0.00	0.92
sin	uniform	cause	n	1000	1.00	1.00	0.00	1.00	1.00	0.00	1.00
sin	uniform	cause	y	200	0.88	0.92	0.24	0.56	0.48	0.36	0.92
sin	uniform	cause	y	1000	0.96	1.00	0.04	0.72	0.84	0.00	1.00
sin	uniform	indep	n	200	1.00	1.00	0.88	0.00	0.92	0.68	0.60
sin	uniform	indep	n	1000	1.00	1.00	1.00	0.04	1.00	1.00	1.00
sin	uniform	indep	y	200	0.80	0.92	0.24	0.04	0.60	0.20	0.68
sin	uniform	indep	y	1000	1.00	1.00	0.20	0.08	0.96	0.16	0.76
sin	normal	homo	n	200	0.00	1.00	1.00	1.00	0.88	0.96	1.00
sin	normal	homo	n	1000	0.00	0.92	1.00	1.00	1.00	1.00	1.00
sin	normal	homo	y	200	0.04	0.64	0.84	1.00	0.56	0.88	1.00
sin	normal	homo	y	1000	0.00	0.32	1.00	1.00	0.88	0.96	1.00
sin	normal	cause	n	200	0.32	1.00	0.24	0.96	0.92	0.28	1.00
sin	normal	cause	n	1000	0.96	1.00	0.28	0.84	1.00	0.72	1.00
sin	normal	cause	y	200	0.20	0.68	0.56	1.00	0.64	0.68	1.00
sin	normal	cause	y	1000	0.56	0.84	0.76	0.96	1.00	0.56	1.00
sin	normal	indep	n	200	0.48	1.00	1.00	0.96	0.96	1.00	1.00
sin	normal	indep	n	1000	0.80	1.00	1.00	0.92	1.00	1.00	1.00
sin	normal	indep	y	200	0.40	0.76	0.84	0.96	0.80	0.80	1.00
sin	normal	indep	y	1000	0.56	0.84	0.88	0.96	1.00	0.80	1.00
sin	bimodal	homo	n	200	0.00	1.00	1.00	1.00	0.88	1.00	0.60
sin	bimodal	homo	n	1000	0.00	1.00	1.00	1.00	1.00	1.00	0.36
sin	bimodal	homo	y	200	0.20	1.00	1.00	1.00	0.48	1.00	0.76
sin	bimodal	homo	y	1000	0.12	1.00	1.00	1.00	0.80	1.00	0.76
sin	bimodal	cause	n	200	0.44	1.00	0.00	1.00	0.72	0.20	0.84
sin	bimodal	cause	n	1000	0.76	1.00	0.00	1.00	1.00	0.84	0.68
sin	bimodal	cause	y	200	0.24	0.96	1.00	1.00	0.76	1.00	0.80
sin	bimodal	cause	y	1000	0.48	1.00	1.00	1.00	0.88	1.00	0.76
sin	bimodal	indep	n	200	0.60	1.00	1.00	1.00	1.00	1.00	0.44
sin	bimodal	indep	n	1000	0.88	1.00	1.00	0.96	1.00	1.00	0.16
sin	bimodal	indep	y	200	0.60	1.00	1.00	1.00	0.48	1.00	0.56
sin	bimodal	indep	y	1000	0.80	1.00	1.00	0.92	0.92	1.00	0.56
sin	discrete5	homo	n	200	0.00	1.00	1.00	1.00	0.00	1.00	1.00
sin	discrete5	homo	n	1000	0.00	1.00	1.00	1.00	0.00	1.00	1.00
sin	discrete5	homo	y	200	0.92	1.00	1.00	1.00	0.76	1.00	0.96
sin	discrete5	homo	y	1000	1.00	1.00	1.00	1.00	0.96	1.00	1.00
sin	discrete5	cause	n	200	0.00	1.00	0.20	1.00	0.00	0.00	1.00
sin	discrete5	cause	n	1000	0.00	1.00	0.00	1.00	0.00	0.12	1.00
sin	discrete5	cause	y	200	0.92	1.00	1.00	1.00	0.60	1.00	1.00
sin	discrete5	cause	y	1000	1.00	1.00	1.00	1.00	0.80	1.00	1.00
sin	discrete5	indep	n	200	0.00	1.00	1.00	0.92	0.00	1.00	1.00
sin	discrete5	indep	n	1000	0.00	1.00	1.00	1.00	0.00	1.00	1.00
sin	discrete5	indep	y	200	1.00	1.00	1.00	0.96	0.72	1.00	0.68
sin	discrete5	indep	y	1000	1.00	1.00	1.00	1.00	0.96	1.00	0.80

Table 16. Accuracy with exact 95% intervals for every rule and collection.

Rule	Tübingen	iid	walks	walks (19)
IGCI	0.56 [0.46,0.66]	0.94 [0.85,1.00]	0.89 [0.71,0.99]	0.96 [0.91,0.99]
RECI	0.61 [0.51,0.70]	0.98 [0.94,1.00]	0.94 [0.86,1.00]	0.83 [0.79,0.87]
ANM-HSIC	0.61 [0.51,0.71]	0.72 [0.61,0.83]	0.63 [0.47,0.81]	0.43 [0.38,0.50]
Pairwise LiNGAM	0.61 [0.51,0.70]	0.78 [0.68,0.87]	0.71 [0.60,0.80]	0.67 [0.62,0.72]
PDM	0.40 [0.30,0.50]	0.93 [0.84,0.99]	0.84 [0.58,0.97]	–
LOCI	0.65 [0.56,0.74]	0.77 [0.66,0.86]	0.68 [0.57,0.84]	–
ANM, Gaussian process	0.55 [0.46,0.65]	0.77 [0.68,0.87]	0.67 [0.55,0.81]	–
Compression, first impl.	0.61 [0.52,0.70]	0.30 [0.19,0.40]	0.61 [0.48,0.69]	–
Compression, faithful	0.67 [0.58,0.75]	0.38 [0.29,0.48]	0.52 [0.45,0.57]	0.50 [0.43,0.57]
TRA	0.65 [0.54,0.75]	0.86 [0.77,0.94]	0.61 [0.40,0.76]	0.56 [0.50,0.61]
AVICI	0.59 [0.49,0.69]	0.90 [0.82,0.97]	0.56 [0.42,0.70]	0.63 [0.58,0.68]
RESIT	0.61 [0.52,0.70]	0.71 [0.61,0.81]	0.65 [0.52,0.79]	0.48 [0.43,0.53]

Rule	Sachs	Bernoulli	Changepoints	Malus
IGCI	0.65 [0.41,0.88]	1.00 [1.00,1.00]	1.00 [1.00,1.00]	0.08 [0.00,0.17]
RECI	0.53 [0.29,0.76]	0.25 [0.00,0.62]	1.00 [1.00,1.00]	0.56 [0.52,0.58]
ANM-HSIC	0.59 [0.35,0.82]	0.31 [0.06,0.62]	0.57 [0.29,0.84]	0.31 [0.27,0.33]
Pairwise LiNGAM	0.53 [0.29,0.76]	0.88 [0.69,1.00]	0.43 [0.24,0.60]	0.40 [0.21,0.58]
PDM	–	–	–	–
LOCI	0.47 [0.24,0.71]	0.56 [0.38,0.75]	0.80 [0.57,0.97]	0.38 [0.29,0.46]
ANM, Gaussian process	0.56 [0.32,0.79]	0.47 [0.22,0.72]	0.54 [0.38,0.66]	0.33 [0.25,0.50]
Compression, first impl.	0.76 [0.53,0.94]	–	–	–
Compression, faithful	0.65 [0.41,0.88]	0.50 [0.19,0.81]	1.00 [1.00,1.00]	0.60 [0.48,0.71]
TRA	0.59 [0.35,0.82]	0.62 [0.38,0.88]	–	0.54 [0.46,0.62]
AVICI	0.53 [0.29,0.76]	0.62 [0.25,1.00]	0.40 [0.17,0.67]	0.02 [0.00,0.06]
RESIT	0.47 [0.29,0.65]	0.25 [0.06,0.44]	0.53 [0.24,0.77]	0.50 [0.50,0.50]

Table 17. The ten reporting universes with at least 30 pairs per rule: weighting, minimum number of rows, coverage (%), pairs in the subset, smallest number of pairs evaluated for any rule, and the accuracy of each rule (a subset of the columns of Table 4; CmpF: faithful compression; the other four columns are in the repository file `results/exp47_multiverse.json`).

Weights	Rows	Cov.	n	min	IGCI	RECI	ANM-H	LiNGAM	LOCI	TRA	CmpF	AVICI	RESIT
official	0	100	102	82	0.66	0.71	0.61	0.62	0.64	0.65	0.75	0.61	0.62
official	0	75	102	62	0.64	0.77	0.58	0.65	0.67	0.71	0.78	0.60	0.65
official	0	50	102	41	0.66	0.87	0.57	0.74	0.63	0.74	0.88	0.58	0.76
official	250	100	76	56	0.71	0.75	0.65	0.54	0.71	0.70	0.75	0.59	0.63
official	250	75	76	42	0.69	0.84	0.63	0.56	0.77	0.74	0.77	0.60	0.67
none	0	100	102	82	0.56	0.61	0.61	0.61	0.65	0.65	0.67	0.59	0.61
none	0	75	102	62	0.50	0.68	0.57	0.63	0.63	0.71	0.72	0.63	0.64
none	0	50	102	41	0.47	0.76	0.57	0.71	0.63	0.76	0.84	0.65	0.75
none	250	100	76	56	0.55	0.67	0.70	0.55	0.72	0.68	0.71	0.54	0.66
none	250	75	76	42	0.51	0.75	0.67	0.58	0.75	0.76	0.77	0.61	0.70

Table 18. Sachs: skeleton F_1 of 22 configurations with a 95% bootstrap interval over the 55 variable pairs.

Configuration	F_1	interval
LRG	0.67	[0.42, 0.85]
BF-BIC+BOSS	0.67	[0.42, 0.85]
PC-GCM α =0.01	0.64	[0.38, 0.83]
PC-GCM α =0.05	0.62	[0.36, 0.81]
BTAD	0.62	[0.35, 0.81]
PC α =0.05	0.58	[0.30, 0.80]
GES-BIC	0.58	[0.30, 0.80]
AVICI	0.56	[0.29, 0.77]
NOTEARS-MLP	0.56	[0.29, 0.77]
PC α =0.01	0.52	[0.24, 0.74]
DirectLiNGAM	0.52	[0.24, 0.74]
PCMCI+	0.52	[0.24, 0.74]
GRaSP	0.52	[0.24, 0.74]
BOSS	0.52	[0.24, 0.74]
CERT-CD nonparametric k=3	0.45	[0.15, 0.69]
DAGMA	0.45	[0.15, 0.69]
RESIT	0.45	[0.24, 0.63]
NOTEARS	0.30	[0.00, 0.56]
GOLEM	0.30	[0.00, 0.56]
DAG-GNN	0.30	[0.00, 0.56]
GraN-DAG	0.11	[0.00, 0.32]
CAM-UV	0.11	[0.00, 0.33]

Table 20. Kendall rank correlation between the orderings of the rules on two collections, with a permutation p-value (5000 permutations of one ordering) in parentheses; rules run on both collections, at least five.

	Tüb.	iid	walks	walks19	Sachs	Bern.	Chgpt.	Malus
Tübingen	–	-0.38 (0.11)	-0.30 (0.21)	-0.42 (0.20)	-0.12 (0.68)	-0.19 (0.54)	+0.24 (0.46)	+0.51 (0.06)
iid	-0.38 (0.11)	–	+0.53 (0.02)	+0.57 (0.06)	-0.17 (0.53)	+0.34 (0.21)	-0.03 (1.00)	-0.16 (0.60)
walks	-0.30 (0.21)	+0.53 (0.02)	–	+0.43 (0.18)	-0.27 (0.30)	+0.07 (0.86)	+0.26 (0.41)	-0.02 (1.00)
walks19	-0.42 (0.20)	+0.57 (0.06)	+0.43 (0.18)	–	+0.04 (1.00)	+0.52 (0.11)	+0.21 (0.65)	-0.07 (0.90)
Sachs	-0.12 (0.68)	-0.17 (0.53)	-0.27 (0.30)	+0.04 (1.00)	–	+0.27 (0.35)	+0.41 (0.18)	-0.02 (1.00)
Bernoulli	-0.19 (0.54)	+0.34 (0.21)	+0.07 (0.86)	+0.52 (0.11)	+0.27 (0.35)	–	-0.06 (0.92)	-0.25 (0.37)
Changepoints	+0.24 (0.46)	-0.03 (1.00)	+0.26 (0.41)	+0.21 (0.65)	+0.41 (0.18)	-0.06 (0.92)	–	+0.26 (0.40)
Malus	+0.51 (0.06)	-0.16 (0.60)	-0.02 (1.00)	-0.07 (0.90)	-0.02 (1.00)	-0.25 (0.37)	+0.26 (0.40)	–